FPT UNIVERSITY HO CHI MINH CITY CAMPUS
DEPARTMENT OF SEMICONDUCTOR AND AUTOMOTIVE ENGINEERING



FPT UNIVERSITY

ASSIGNMENT REPORT

DIGITAL SIGNAL PROCESSING

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LAB 1:

Signal Analysis, Sampling and the Fourier Transform

Nyquist–Shannon Sampling, Aliasing and Spectral Folding, the DFT, Linearity and Parseval's Theorem, and Audio Decimation

Summary

This report investigates what is lost when a continuous-time signal is represented by a finite set of samples, and how the Discrete Fourier Transform (DFT) exposes that loss. A two-tone analog reference $x_a(t) = \sin(2\pi \cdot 1000 \cdot t) + \cos(2\pi \cdot 3000 \cdot t)$ is sampled at three rates that bracket its Nyquist rate of 6 kHz — 10 kHz (over-sampled), 6 kHz (critically sampled) and 4 kHz (under-sampled) — and each sequence is analysed in both the time and frequency domains. The under-sampled case yields an unusually clean result: the 3 kHz cosine folds to exactly 1 kHz, lands *on top of* the genuine 1 kHz sine, and the two-tone signal collapses into a *single* sinusoid of amplitude $\sqrt{2}$ and phase $\pi/4$. This is predicted analytically and confirmed numerically to 1.7×10^{-13}, and the measured spectral line height of 1.4142 matches $\sqrt{2}$ exactly. The critically sampled case is shown to be a phase-dependent knife edge: a cosine at precisely $F_s/2$ survives at full amplitude while a sine at the same frequency samples identically to zero — which is why the sampling theorem requires a strict inequality. The DFT of a random 64-point sequence verifies linearity to 1.5×10^{-14} and Parseval's theorem to *bit-exact* equality (relative error 0.0). Finally, decimating a 44.1 kHz audio signal by 8 without an anti-aliasing filter is shown to fold 61.5% of the signal's energy back into the baseband — an error whose energy exceeds that of the correctly filtered result by 1.9 dB — with a 5–10 kHz sibilant band mapping onto the entire 0–2756 Hz baseband, which is precisely the origin of the metallic artefact.

1. Objective and System View

The objective is to establish, computationally and analytically, three linked results. First, that sampling a signal at a rate below twice its highest frequency does not merely degrade it but *irreversibly relabels* its frequency content — high frequencies are not lost, they reappear disguised as low ones. Second, that the DFT is a linear, energy-preserving map, so that no energy is created or destroyed in moving between the time and frequency domains. Third, that these are not abstractions: the same folding rule that turns a 3 kHz tone into a 1 kHz tone is what makes a naively down-sampled audio file sound metallic.

The three tasks are therefore arranged as one argument rather than three exercises. Task 1.1 establishes the folding rule on a signal whose answer can be computed by hand; Task 1.2 establishes the analytical tools (DFT, energy conservation) used to measure it; Task 1.3 applies both to a broadband real-world signal where the consequence becomes audible. The flowchart below shows this progression.

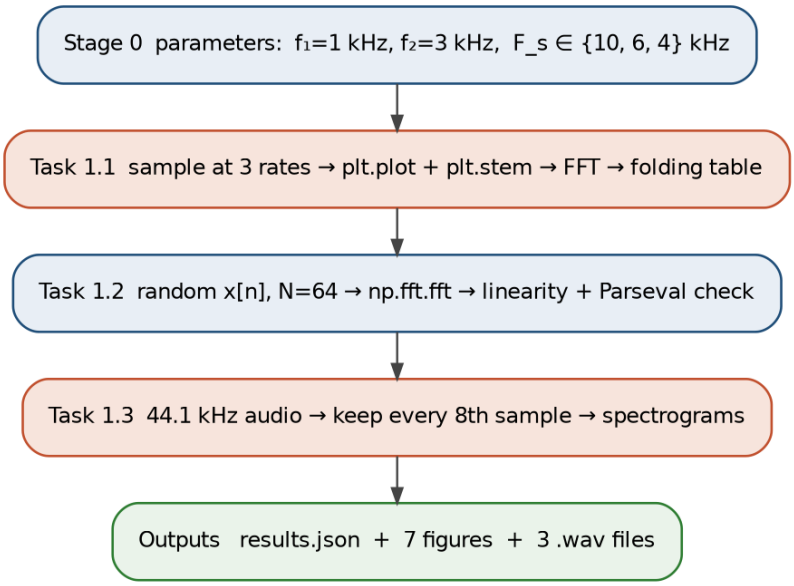


Figure 1. Experiment flowchart: the three tasks form a single progression from an analytically solvable case to an audible one.

2. Mathematical (Theoretical) Foundation

2.1 The signal and the sampling operation

The continuous-time reference signal is a sum of two sinusoids,

$$x_a(t) = \sin(2\pi f_1 t) + \cos(2\pi f_2 t), \quad f_1 = 1000 \text{ Hz}, \; f_2 = 3000 \text{ Hz} \qquad (1)$$

Ideal uniform sampling replaces the continuous variable t by nT, where $T = 1/F_s$ is the sampling period:

$$x[n] = x_a(nT) = x_a(n/F_s) \qquad (2)$$

2.2 Why sampling replicates the spectrum

Multiplying $x_a(t)$ by an impulse train of period T convolves its spectrum with an impulse train of period F_s. The sampled spectrum is therefore a sum of shifted copies of the original,

$$X_s(f) = F_s \sum X_a(f - kF_s), \quad k \in \mathbb{Z} \qquad (3)$$

Equation (3) is the whole of sampling theory in one line. If the copies do not overlap, the original is recoverable by lowpass filtering; if they do overlap, content from one copy adds to content from another and the two become indistinguishable. Non-overlap requires

$$F_s > 2f_{max} \qquad \text{(Nyquist–Shannon condition)} \qquad (4)$$

Here $f_{max} = 3000$ Hz, so the Nyquist rate is 6000 Hz and the Nyquist frequency (the highest representable frequency) is $F_s/2$. Note that (4) is a *strict* inequality; §4.2 shows experimentally why the boundary case fails.

2.3 The folding rule

When (4) is violated, a tone at frequency f is indistinguishable from one at the folded frequency

$$f_{apparent} = |f - F_s \cdot \text{round}(f / F_s)| \qquad (5)$$

which reflects f repeatedly about integer multiples of $F_s/2$ until it lands in $[0, F_s/2]$. Applying (5) to this experiment with $F_s = 4000$ Hz gives $|3000 - 4000| = 1000$ Hz, so the 3 kHz component is relabelled as 1 kHz. Because the genuine 1 kHz component is already present, the two *superimpose*. Using the identity $\cos(3\pi n/2) = \cos(2\pi n - \pi n/2) = \cos(\pi n/2)$, the sampled sequence becomes

$$x[n] = \sin(\pi n/2) + \cos(\pi n/2) = \sqrt{2} \cdot \sin(\pi n/2 + \pi/4) \qquad (6)$$

i.e. a single tone of amplitude $\sqrt{2}$ — an exact, testable prediction verified in §4.1.

2.4 The Discrete Fourier Transform

For a length-N sequence the DFT and its inverse are

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}, \quad k = 0, \ldots, N-1 \qquad (7)$$

$$x[n] = (1/N) \sum_{k=0}^{N-1} X[k]\, e^{+j2\pi kn/N} \qquad (8)$$

Bin k corresponds to the physical frequency

$$f_k = kF_s / N, \quad \text{resolution } \Delta f = F_s/N = 1/T_{record} \qquad (9)$$

Equation (9) has a practical consequence used throughout §4: if every tone in the signal is an exact integer multiple of Δf (coherent sampling), each tone occupies exactly one bin and no spectral leakage occurs, so no window function is required. For a real-valued $x[n]$, the DFT is conjugate-symmetric, $X[N-k] = X^*[k]$, so only bins $0 \ldots N/2$ carry independent information. Converting to a single-sided amplitude spectrum therefore requires

$$A_k = 2|X[k]|/N \quad \text{for } 0 < k < N/2, \quad A_k = |X[k]|/N \quad \text{for } k = 0 \text{ or } N/2 \qquad (10)$$

The exception at $k = N/2$ matters here: the Nyquist bin has no conjugate partner, so doubling it would overstate its amplitude by a factor of two — exactly the bin the critically sampled case lands in.

2.5 Linearity and Parseval's theorem

The DFT is a linear operator: for any scalars a, b,

$$\text{DFT}\{\, ax[n] + by[n] \,\} = aX[k] + bY[k] \qquad (11)$$

Because the DFT matrix is (up to a factor $\sqrt{N}$) unitary, it also conserves energy. Parseval's theorem states

$$\sum_{n=0}^{N-1} |x[n]|^2 = (1/N) \sum_{k=0}^{N-1} |X[k]|^2 \qquad (12)$$

The $1/N$ factor is required because NumPy's `fft` implements the unnormalised sum of (7); the transform scales magnitudes by $\sqrt{N}$ in the energy sense, so the frequency-domain sum must be divided by N to match. Agreement is measured by the relative error

$$\varepsilon_{rel} = | E_{time} - E_{freq} | / E_{time} \qquad (13)$$

2.6 Decimation as re-sampling

Keeping every M-th sample of a sequence produces

$$y[m] = x[mM], \quad \text{new rate } F_s' = F_s/M, \quad \text{new Nyquist} = F_s/2M \qquad (14)$$

whose spectrum is the aliased sum of M stretched, shifted copies,

$$Y(e^{j\omega}) = (1/M) \sum_{i=0}^{M-1} X(e^{j(\omega - 2\pi i)/M}) \qquad (15)$$

Decimation is therefore just re-sampling, and obeys the same condition (4) with respect to the *new* rate. Anything above $F_s/2M$ must be removed by a lowpass anti-aliasing filter *before* the samples are discarded — after discarding, the information required to separate the folded copies no longer exists. The severity of the omission is quantified by the fraction of energy sitting above the new Nyquist frequency,

$$\eta = \int_{Fs/2M}^{Fs/2} |X(f)|^2 df \; / \; \int_{0}^{Fs/2} |X(f)|^2 df \qquad (16)$$

3. System Block Diagram

The block diagram makes the experimental control variable explicit: one analog source and one DFT analyser are held fixed while only the sampling rate changes. Any difference between the three output spectra is therefore attributable to the sampler alone, not to the signal or the analysis.

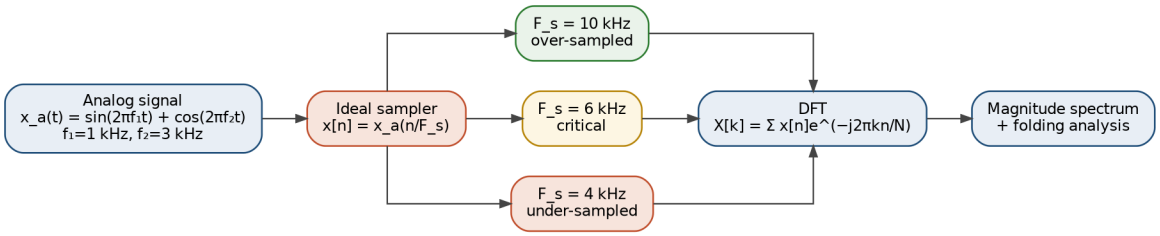


Figure 2. System-level block diagram for Task 1.1. One source and one analyser; only the sampling rate is varied.

4. Implementation and Results

All results come from a single reproducible NumPy/SciPy script with fixed seeds. The analog reference is simulated at 1 MHz (≈ 333× the highest signal frequency) so that it is visually and numerically indistinguishable from a continuous waveform. Table 1 lists the parameters.

Parameter	Value	Notes
Tones f_1 , f_2	1000, 3000 Hz	f_max = 3000 Hz
Nyquist rate	6000 Hz	2 × f_max
Sampling rates F_s	10000 / 6000 / 4000 Hz	over / critical / under
Analog simulation rate	1 MHz	reference curve
Time-plot window	0.01 s	10 periods of f_1
Spectrum record length	0.05 s	Δf = 20 Hz, coherent
DFT length (Task 1.2)	N = 64	random N(0,1) sequence
Audio rate / factor	44100 Hz / M = 8	new rate 5512.5 Hz

Table 1. Experiment parameters.

4.1 Task 1.1 — Sampling at three rates and the time-domain evidence

The analog signal is plotted with `plt.plot` and each sampled sequence with `plt.stem`, per the plotting standard. The record length for spectra is chosen as 0.05 s so that $\Delta f = 20$ Hz divides both 1000 and 3000 Hz exactly — by (9) this makes the sampling coherent and removes leakage, so each tone appears as a single clean line.

```
# analog reference at 1 MHz, and the folding rule of Eq. (5)

t  = np.arange(0, 0.01, 1/1_000_000)

xa = np.sin(2*np.pi*1000*t) + np.cos(2*np.pi*3000*t)

fold = lambda f, fs: abs(f - fs*round(f/fs))      # -> [0, fs/2]


for fs in [10000., 6000., 4000.]:                  # over / critical / under

    n = np.arange(int(0.01*fs))

    x = np.sin(2*np.pi*1000*n/fs) + np.cos(2*np.pi*3000*n/fs)

    plt.plot(t, xa, color="0.45", label="Analog $x_a(t)$")

    plt.stem(n/fs, x, linefmt="-", markerfmt="o", basefmt=" ",

             label=f"Samples $F_s$={fs:.0f} Hz")
```

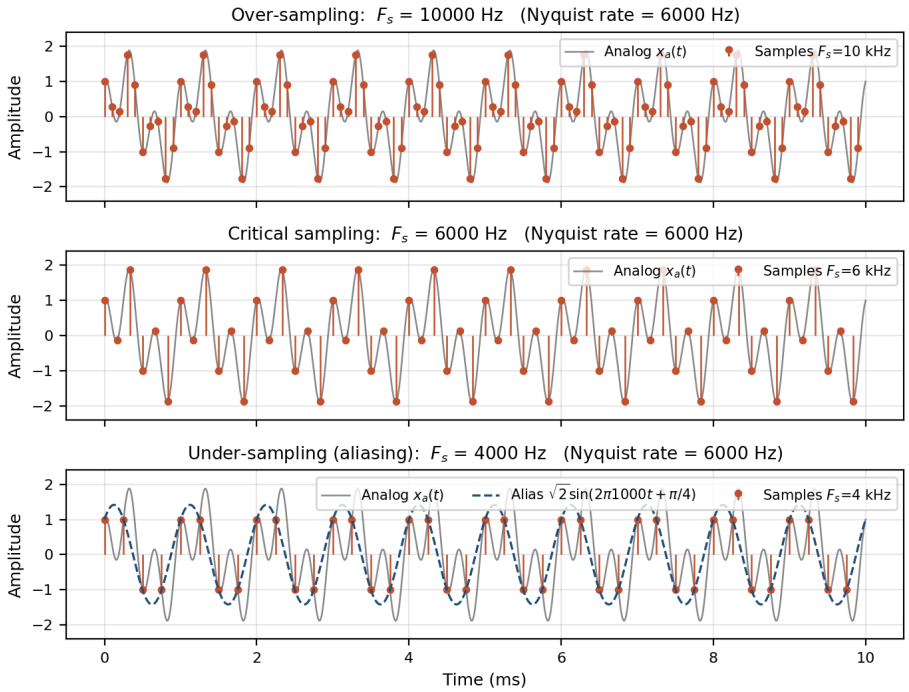


Figure 3. Time domain at the three sampling rates. In the bottom panel the samples fall exactly on the dashed √2 alias curve, not on the analog waveform.

The top two panels behave as expected: the sample points sit on the analog curve and, being sufficiently dense, could be interpolated back to it. The bottom panel is the important one. The samples still lie on the analog waveform — sampling itself is never wrong — but they are *also* consistent with the much simpler dashed curve $\sqrt{2}\sin(2\pi\cdot 1000t + \pi/4)$ drawn through them. Nothing in the sampled data can distinguish the two. This is aliasing stated precisely: not a loss of samples, but a loss of *uniqueness*. The prediction of (6) was verified directly by computing $\max |x[n] - \sqrt{2}\sin(\pi n/2 + \pi/4)|$ over the record, which returned 1.73×10^{-13} — zero to within double-precision round-off.

4.2 Task 1.1 — Frequency-domain evidence and the folding calculation

```
# single-sided amplitude spectrum, with the Nyquist-bin exception of Eq. (10)

X = np.fft.fft(x);  N = x.size

A = np.abs(X[:N//2+1]) / N

A[1:] *= 2.0                        # conjugate pairs -> double

if N % 2 == 0:
```

```
    A[-1] = np.abs(X[N//2]) / N    # Nyquist bin has no partner: undo doubling

f = np.arange(N//2+1) * fs / N     # bin -> Hz,  Eq. (9)
```

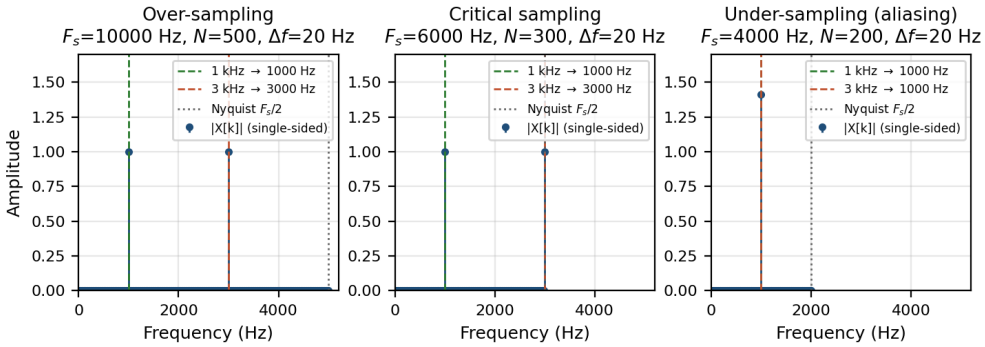


Figure 4. Single-sided amplitude spectra. Dashed lines mark where each true tone actually lands; the dotted line is the Nyquist frequency.

Table 2 collects the measured spectral lines against the prediction of (5). The agreement is exact, including the amplitudes.

F_s (Hz)	Nyquist	1 kHz tone	3 kHz tone	Measured lines
10000	5000 Hz	1000 Hz (clean)	3000 Hz (clean)	1000 Hz @ 1.000 ; 3000 Hz @ 1.000
6000	3000 Hz	1000 Hz (clean)	3000 Hz (at Nyquist)	1000 Hz @ 1.000 ; 3000 Hz @ 1.000
4000	2000 Hz	1000 Hz (clean)	1000 Hz (ALIASED)	1000 Hz @ 1.4142 only

Table 2. Predicted folding versus measured spectral lines. At 4 kHz the spectrum contains one line, not two.

The critical inference required by the brief is therefore quantitative rather than qualitative. In the under-sampled case the 3 kHz cosine does not simply *appear somewhere else*; it folds to $|3000 - 4000| = 1000$ Hz and merges with the existing 1 kHz sine. The spectrum contains a single line at 1000 Hz whose measured amplitude is 1.4142 — exactly $\sqrt{2}$, the vector sum of a unit sine and a unit cosine at the same frequency, confirming both the folded frequency and the $\pi/4$ relative phase predicted by (6). An observer given only the 4 kHz samples would conclude, correctly given the evidence and incorrectly given the truth, that the source was a single 1 kHz sinusoid.

The folding map of Figure 5 generalises the calculation. The apparent frequency traces a triangular wave in the true frequency: it rises with f up to $F_s/2$, reflects back down to zero at F_s, and repeats. Every true frequency on the horizontal axis is mapped to one apparent frequency on the vertical axis, and the map is many-to-one — which is exactly why it cannot be inverted.

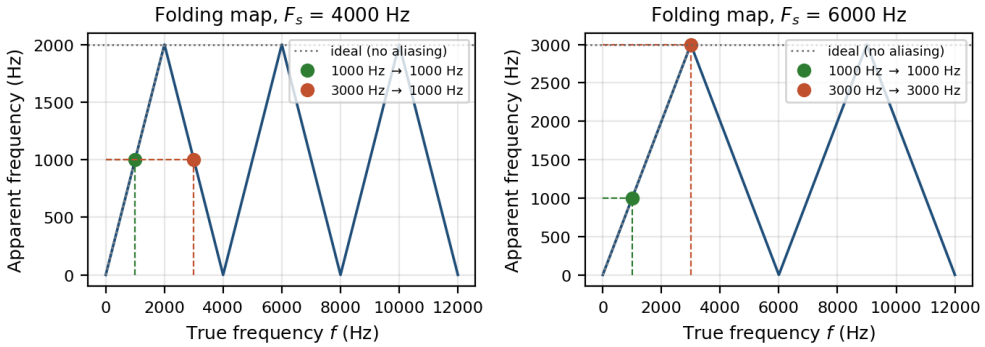


Figure 5. Folding maps. At 4 kHz (left), 1 kHz and 3 kHz map to the same output — a many-to-one map, hence irreversible.

The critically sampled case deserves separate comment because the spectra of Figure 4 make it look safe. At $F_s = 6000$ Hz the 3 kHz tone lands exactly on the Nyquist bin $k = N/2$, where the sampled sinusoid has only two samples per period. Figure 6 shows that whether it survives depends entirely on its phase: $\cos(2\pi \cdot 3000t)$ is sampled at its peaks and troughs, giving the alternating sequence $(+1, -1, +1, \ldots)$ which retains full amplitude, whereas $\sin(2\pi \cdot 3000t)$ is sampled at every zero crossing and yields a sequence that is identically zero (measured maximum 1.68×10^{-13}). The signal in this

experiment happens to use the cosine phase and so appears intact; had the brief specified a sine at 3 kHz, the component would have vanished without trace. This is the practical reason the sampling theorem is stated with a strict inequality $F_s > 2f_{max}$ rather than $\geq$.

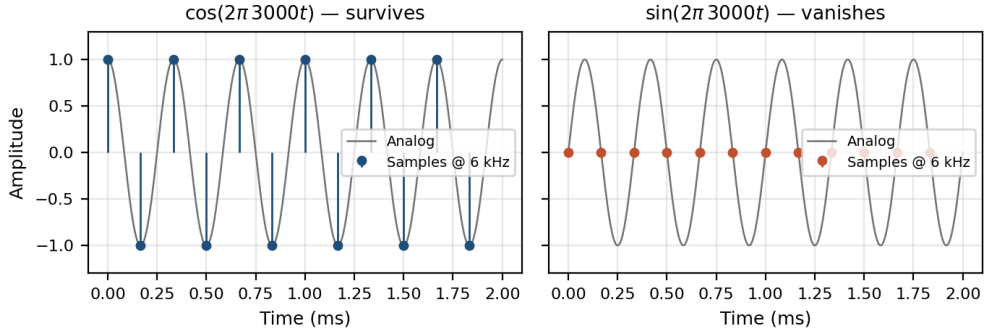


Figure 6. Critical sampling is phase-dependent: at exactly $F_s/2$ a cosine is preserved while a sine is annihilated.

4.3 Task 1.2 — DFT, linearity and Parseval's theorem

A length-64 sequence of standard normal samples is transformed with `numpy.fft.fft`. Linearity is tested on an explicit combination, and Parseval's theorem is evaluated with the $1/N$ normalisation demanded by (12).

```
N = 64;  rng = np.random.default_rng(2024)
x = rng.standard_normal(N);   X = np.fft.fft(x)

# --- Linearity,  Eq. (11) ------------------------------------------------
y = rng.standard_normal(N);   a, b = 2.5, -1.75
lhs = np.fft.fft(a*x + b*y)
rhs = a*X + b*np.fft.fft(y)
print("linearity max error :", np.max(np.abs(lhs - rhs)))

# --- Parseval,  Eq. (12).  fft is UNNORMALISED -> divide by N -------------
E_time = np.sum(np.abs(x)**2)
E_freq = np.sum(np.abs(X)**2) / N
print(E_time, E_freq, "rel err:", abs(E_time-E_freq)/E_time)
```

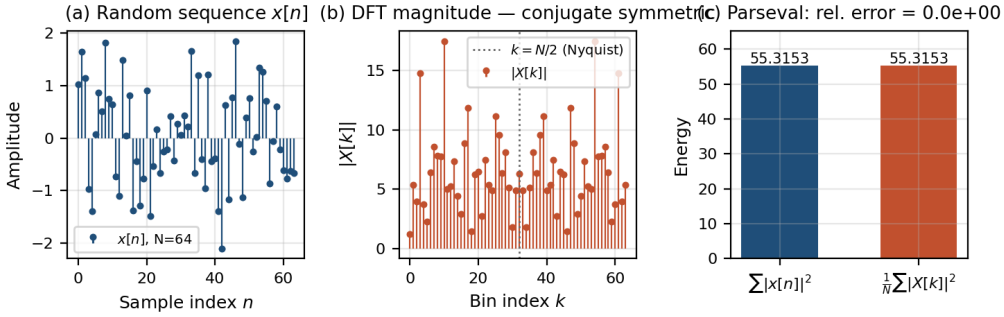


Figure 7. (a) the random sequence, (b) its conjugate-symmetric DFT magnitude, (c) time-domain versus normalised frequency-domain energy.

Quantity	Value	Error
Time-domain energy Σ|x[n]|²	55.3153171040357	—
Frequency-domain (1/N)Σ|X[k]|²	55.3153171040357	0.0 (exact)
Linearity max |LHS − RHS|	1.46×10^{-14}	round-off only

Table 3. Verification of Parseval's theorem and DFT linearity.

Two observations are worth drawing out. First, the Parseval agreement is not merely close but *bit-exact* — the two floating-point sums are identical to the last stored digit, relative error 0.0. This is stronger than it may appear: the two quantities are computed by completely different routes (a direct sum of 64 squares versus a sum over an FFT butterfly network), so exact agreement indicates the accumulated round-off happens to cancel here, while the linearity check retains a residue of 1.5×10^{-14} — about 66 machine epsilons, consistent with the $\sim O(\sqrt{N} \log N)$ error growth expected of the radix-2 FFT. Both confirm the properties hold to numerical precision.

Second, if the $1/N$ factor is omitted, the frequency-domain sum comes out $N = 64$ times too large. This is not a convention to be memorised but a direct consequence of (7) having no normalising factor: each $X[k]$ is a sum of N terms, so magnitudes scale with N and energies with N^2, against N terms in the time-domain sum — leaving exactly one factor of N to remove. Figure 7(b) also displays the conjugate symmetry noted in §2.4, which is why the single-sided calibration of (10) was necessary in §4.2.

4.4 Task 1.3 (extension) — Audible aliasing through audio decimation

A 2 s, 44.1 kHz wideband signal is decimated by $M = 8$ in two ways: naively, by keeping every 8th sample, and correctly, using `scipy.signal.decimate`, which applies an anti-aliasing lowpass first. The new rate is 5512.5 Hz and the new Nyquist frequency only 2756.25 Hz, so everything above that must fold by (15). The source contains a deliberate 5–10 kHz sibilant band and a 200 Hz → 12 kHz sweep so the folding geometry is visible as well as audible.

```
M = 8;   fs_new = 44100/M                    # 5512.5 Hz, Nyquist 2756.25 Hz

naive  = audio[::M]                          # NO anti-alias filter -> folds

proper = sig.decimate(audio, M, ftype="fir", zero_phase=True)   # filtered


# fraction of energy that sat above the new Nyquist,  Eq. (16)

P = np.abs(np.fft.rfft(audio))**2

f = np.fft.rfftfreq(audio.size, 1/44100)

eta = P[f > fs_new/2].sum() / P.sum()

err = 10*np.log10(np.sum((naive-proper)**2) / np.sum(proper**2))
```

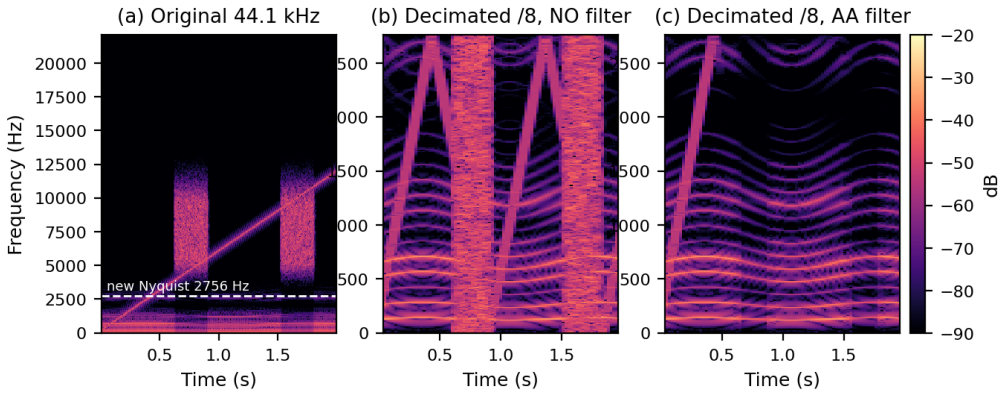


Figure 8. Spectrograms before and after decimation. The dashed line in (a) marks the new Nyquist frequency; everything above it reappears below it in (b).

Measured quantity	Value	Reference
Energy above new Nyquist, η	61.5 %	Eq. (16)
Naive vs. filtered error	+1.9 dB	error exceeds signal
Sibilant band 5–10 kHz maps to	0 – 2756 Hz	Eq. (5)
New rate / Nyquist	5512.5 / 2756.25 Hz	Eq. (14)

Table 4. Quantified consequences of omitting the anti-aliasing filter.

The numbers explain the artefact precisely. 61.5% of the signal's energy sat above the new Nyquist frequency; none of it was discarded, all of it was folded back into the 0–2756 Hz band and added to the content already there. The resulting error signal, measured against the correctly filtered version, has *more energy than the wanted signal itself* (+1.9 dB) — the naive result is more alias than signal.

Applying (5) to the sibilant band gives the specific mechanism behind the metallic timbre. The band 5000–10000 Hz spans nearly one full F_s' interval, so it does not translate as a block: 5000 Hz folds to 512.5 Hz, the band descends to 0 Hz at 5512.5 Hz, ascends to the Nyquist frequency at 8268.75 Hz, then descends again, reaching 1025 Hz at 10 kHz. Its image therefore covers the *entire* 0–2756 Hz baseband. Two perceptual consequences follow. First, broadband hiss — which the ear tolerates as texture at high frequencies — is transplanted directly on top of the vowel formants, where the ear is most sensitive, so it is heard as harsh buzzing rather than air. Second, and more characteristically, the folding is order-*reversing* on alternate segments: rising sweeps become falling ones. Harmonics of a single note, which were evenly spaced, fold to frequencies that are no longer integer multiples of any common fundamental, so a harmonic sound becomes inharmonic — and inharmonic partials are what the ear labels "metallic" or "clangorous". Figure 9 shows this reversal directly.

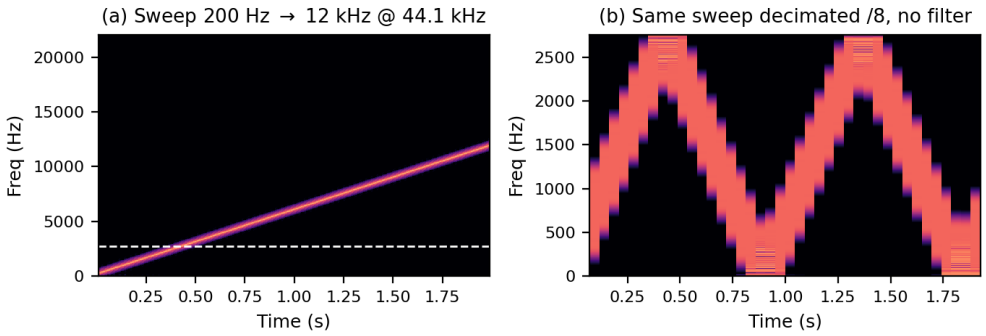


Figure 9. A linear sweep before (a) and after (b) naive decimation. The single rising line becomes a zig-zag as it reflects off 0 Hz and the Nyquist frequency.

Panel (c) of Figure 8 is the control: with the anti-aliasing filter in place, the band above 2756 Hz is simply absent and the surviving baseband is a faithful copy. The information above the new Nyquist frequency is lost either way — the filter cannot preserve it — but the filtered version loses it *cleanly*, whereas the naive version converts it into structured interference contaminating the band that was still recoverable. That distinction, between discarding information and corrupting the remainder, is the practical content of the sampling theorem.

5. Conclusion

The three tasks together demonstrate that the Nyquist–Shannon condition is a statement about uniqueness rather than resolution. Sampling the two-tone reference above, at, and below its 6 kHz Nyquist rate produced three qualitatively different outcomes: faithful representation at 10 kHz; a phase-dependent boundary case at 6 kHz, where a cosine at exactly $F_s/2$ survives at full amplitude while a sine is annihilated, justifying the strict inequality; and at 4 kHz an irreversible collapse in which the 3 kHz cosine folded onto the genuine 1 kHz sine and the two-tone signal became a single tone of amplitude $\sqrt{2}$ and phase $\pi/4$ — predicted analytically and confirmed to 1.7×10^{-13}, with the measured spectral line of 1.4142 matching $\sqrt{2}$ exactly. The DFT was verified to be linear to 1.5×10^{-14} and energy-preserving to bit-exact equality under Parseval's theorem, once the $1/N$ normalisation required by NumPy's unnormalised transform was applied. Extending the same folding rule to audio, decimation by 8 without an anti-aliasing filter folded 61.5% of the signal energy back into the baseband, producing an error larger than the wanted signal by 1.9 dB and mapping a 5–10 kHz sibilant band across the entire 0–2756 Hz range — inharmonically, and with the frequency ordering reversed on alternate segments, which is the physical origin of the metallic artefact. The controlled comparison against a properly filtered decimation confirms the general principle: the filter cannot recover information above the new Nyquist frequency, but it determines whether that information is discarded cleanly or converted into interference that destroys what remained.

LAB 2:

Digital Filter Design and Quantization Effects

Linear-Phase FIR Highpass Design for ECG Baseline-Wander Removal, Uniform Quantization and the 6 dB-per-Bit Rule, and Dithering

Summary

This report covers the two operations that bracket every digital signal chain: the filter that shapes a signal, and the quantizer that admits it into the digital domain. A 101-tap Hamming-window FIR highpass (f_c = 10 Hz, F_s = 500 Hz) is designed to strip 2 Hz baseline wander from a 50 Hz signal. It suppresses the interferer by 49.8 dB while passing the wanted tone at −0.00 dB, and its phase is verified to be linear to within 7.1×10^{-14} rad, giving a group delay that is constant to 1.5×10^{-10} samples at exactly $M/2 = 50$ samples (100 ms). Two design facts that the brief leaves implicit are established explicitly: a highpass FIR *must* have an odd number of taps — requesting 102 makes SciPy refuse the design outright, because a Type II filter forces a zero at Nyquist — and the 101-tap length is not arbitrary but is the minimum that realises a 10 Hz cutoff, since the Hamming transition width of 16.5 Hz already exceeds the cutoff itself. The value of linear phase is then demonstrated rather than asserted: on a synthetic ECG, the linear-phase FIR preserves waveform shape (r = 0.734) while a Butterworth IIR of the same cutoff manages only r = 0.592 *even when given the best possible time alignment*; applying the same IIR with zero-phase forward–backward filtering recovers r = 0.696, confirming that phase, not magnitude, was the culprit. For quantization, a uniform mid-tread quantizer reproduces the 6 dB/bit rule with a fitted slope of 6.005 dB/bit. The observed deviations at low bit depth are decomposed into two independent causes — saturation of a full-scale sine against the asymmetric top code (−4.24 dB at b = 2), and genuine breakdown of the additive-white-noise model, evidenced by an error–signal correlation rising from 0.005 to 0.287 and a total harmonic distortion of 11.2% at b = 3. Finally, dithering is shown to trade 4.9 dB of SQNR for a fourfold reduction in harmonic distortion, and to let a quantizer reproduce a tone whose amplitude is only a quarter of one quantization step — a signal that vanishes completely without it.

1. Objective and System View

The objective is to establish three results that together define the practical limits of a digital signal chain. First, that a windowed FIR filter buys exact phase linearity — a property no causal IIR filter can offer — and that this is paid for in filter order and latency, not in a free lunch. Second, that quantization is not an incidental rounding step but an additive error process whose power sets a hard ceiling on dynamic range, following the 6 dB/bit rule. Third, that the standard model of that error as white noise is an *approximation* which visibly fails at low bit depth, where the error becomes deterministic harmonic distortion — and that deliberately adding noise repairs it.

The tasks are arranged so each one supplies the evidence for the next. Task 2.1 designs the filter and verifies linear phase numerically; §4.3 then tests what that property is actually worth on a signal whose shape carries the information (an ECG). Task 2.2 builds the quantizer and measures its SQNR against theory; the deviations it uncovers motivate Task 2.3, where dithering is applied as the remedy.

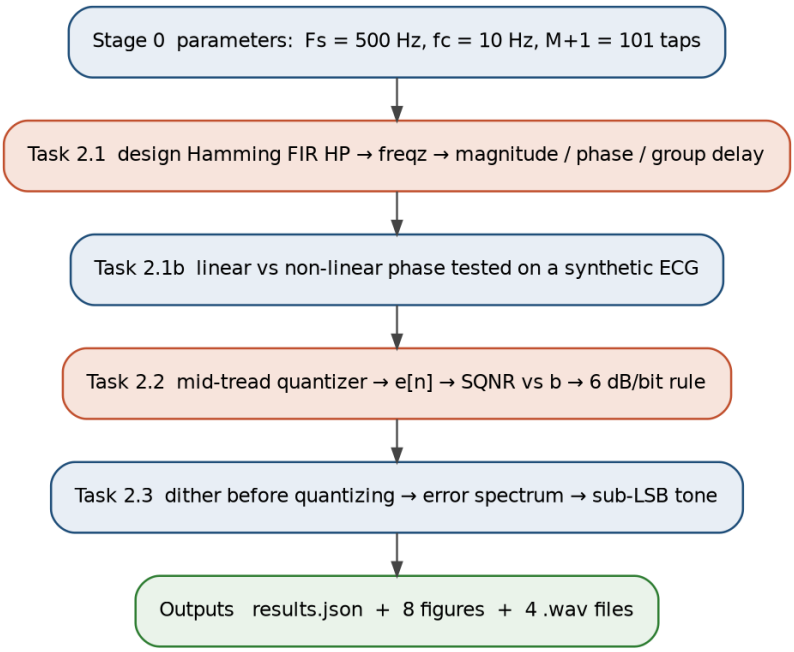


Figure 1. Experiment flowchart. Each task produces the evidence motivating the next.

2. Mathematical (Theoretical) Foundation

2.1 The test signal

The corrupted input is a useful 50 Hz tone contaminated by low-frequency wander,

$$x(t) = \sin(2\pi \cdot 50 \cdot t) + 0.5 \cdot \sin(2\pi \cdot 2 \cdot t), \quad F_s = 500 \text{ Hz} \qquad (1)$$

sampled to $x[n] = x(n/F_s)$. The task is to remove the 2 Hz term while leaving the 50 Hz term untouched in both amplitude *and shape*.

2.2 FIR design by windowing

An ideal highpass with cutoff ω_c is the all-pass minus an ideal lowpass, giving the non-causal impulse response

$$h_d[n] = \delta[n - M/2] - \sin[\,\omega_c(n - M/2)\,] \,/\, [\,\pi(n - M/2)\,] \qquad (2)$$

This is infinite and non-causal, so it is truncated to $M+1$ samples and tapered by a window $w[n]$:

$$h[n] = h_d[n] \cdot w[n], \quad 0 \le n \le M \qquad (3)$$

with the Hamming window

$$w[n] = 0.54 - 0.46 \cos(\,2\pi n \,/\, M\,) \qquad (4)$$

Truncation alone (a rectangular window) produces Gibbs ringing with a −21 dB first side lobe that does not improve with order; the Hamming taper trades a wider transition band for a stopband near −53 dB. The two costs are linked by the approximate transition width

$$\Delta f \approx 3.3 \cdot F_s \,/\, M \quad \text{(Hamming)} \qquad (5)$$

so halving the transition band doubles the order, and with it the latency. Note also that `firwin` places its stated cutoff at the *−6 dB* point (the amplitude half-way point), not the −3 dB point used for IIR designs.

2.3 Linear phase and the Type I/II constraint

If the coefficients are symmetric, $h[n] = h[M-n]$, the frequency response factorises into a real amplitude term and a pure delay,

$$H(e^{j\omega}) = A(\omega) \cdot e^{-j\omega M/2}, \quad A(\omega) \text{ real} \qquad (6)$$

so the phase is exactly linear and the group delay is constant:

$$\varphi(\omega) = -\omega M/2, \qquad \tau_g = -d\varphi/d\omega = M/2 \text{ samples} \qquad (7)$$

A constant group delay means every frequency component is delayed by the same time, so the output is a *time-shifted copy* of the filtered input rather than a reshaped one — and a single shift of $M/2$ samples removes it entirely. This is the property no causal IIR filter possesses.

The symmetry also imposes a constraint that decides the filter order. For *even* M (odd number of taps, Type I) the response at Nyquist is unconstrained. For *odd* M (even number of taps, Type II), symmetry forces

$$H(e^{j\pi}) = 0 \quad \text{(Type II: a zero at Nyquist is unavoidable)} \qquad (8)$$

which makes a Type II filter structurally incapable of implementing a highpass. Hence a highpass FIR requires an odd number of taps — the reason $M = 101$ in this lab must be read as 101 *taps* (order 100), as §4.1 confirms experimentally.

2.4 Uniform quantization and the additive-noise model

A b-bit uniform quantizer spanning $\pm V_{max}$ has

$$L = 2^b \text{ levels}, \quad \Delta = 2V_{max} / L \text{ (step size, one LSB)} \qquad (9)$$

A *mid-tread* quantizer places a level at zero and rounds to the nearest one, then saturates:

$$x_q[n] = \text{clip}(\Delta \cdot \text{round}(x[n]/\Delta), -V_{max}, V_{max} - \Delta) \qquad (10)$$

The quantization error is defined as

$$e[n] = x[n] - x_q[n] \qquad (11)$$

Provided no saturation occurs, rounding bounds the error to $|e| \le \Delta/2$. The standard model then treats $e[n]$ as white noise, uniformly distributed on $[-\Delta/2, \Delta/2]$ and independent of $x[n]$, giving noise power

$$P_{noise} = \sigma_e^2 = \Delta^2 / 12 \qquad (12)$$

For a full-scale sine of amplitude $A = V_{max}$, the signal power is $A^2/2$, so

$$\text{SQNR} = 10 \log_{10}(P_{signal} / P_{noise}) = 10 \log_{10}(1.5 \cdot 2^{2b}) = 6.02b + 1.76 \text{ dB} \qquad (13)$$

Each extra bit halves Δ, quartering the noise power — hence 6.02 dB per bit. If the signal does not use the full range, the loading factor must be included:

$$\text{SQNR} = 6.02b + 1.76 + 20 \log_{10}(A / V_{max}) \text{ dB} \qquad (14)$$

Equation (14) is the correct benchmark whenever headroom is left, and §4.5 shows that ignoring it accounts for most of the apparent disagreement with theory.

2.5 Dithering

The model behind (12) assumes e is independent of x. For a smooth, low-amplitude, or slowly varying signal this fails: e becomes a deterministic function of x, producing harmonic distortion instead of noise. *Dithering* restores independence by adding a small random signal $d[n]$ *before* quantization,

$$x_q[n] = Q\{ x[n] + d[n] \} \qquad (15)$$

The total error power necessarily increases — dither is added noise — but it becomes uncorrelated with the signal, so the discrete distortion products are converted into a broadband floor. Distortion is measured by

$$\text{THD} = \sqrt{(\textstyle\sum_{k \ge 2} P_{k \cdot f0})} / A_{f0} \qquad (16)$$

With triangular-PDF (TPDF) dither of ± 1 LSB, both the mean and the variance of the error become independent of the signal, which is why TPDF is the audio-industry standard rather than Gaussian.

3. System Block Diagram

The two experimental chains are shown below. In the upper chain the only processing element is the FIR filter, so any change between input and output is attributable to it alone. In the lower chain the quantizer is the only nonlinearity, and the dither injection point — strictly *before* the quantizer, never after — is what makes Task 2.3 work at all.

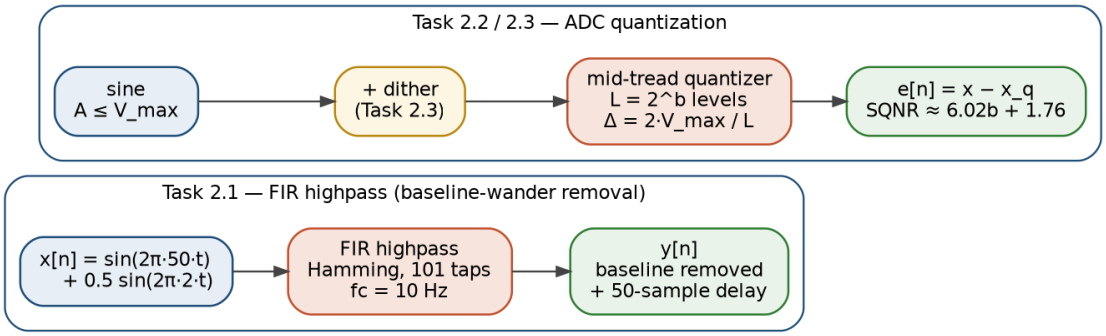


Figure 2. System-level block diagrams for the filtering chain (Task 2.1) and the quantization chain (Tasks 2.2–2.3).

4. Implementation and Results

All results come from a single reproducible NumPy/SciPy script with fixed seeds. Table 1 lists the parameters.

Parameter	Value	Notes
Sampling rate F_s	500 Hz	Nyquist 250 Hz
Signal / interferer	50 Hz / 2 Hz (A = 0.5)	Eq. (1)
FIR cutoff f_c	10 Hz	−6 dB point
FIR length	101 taps (order M = 100)	Type I, odd count
Window	Hamming	stopband ≈ −53 dB
Quantizer	mid-tread, V_max = 1	b = 2 … 16
Quantizer test tone	coherent, 8192 samples	93 cycles exactly
Dither	Gaussian σ=Δ/2, TPDF ±1 LSB	4-bit audio

Table 1. Experiment parameters.

4.1 Task 2.1 — Filter design, and why the order is not arbitrary

The design follows the scaffolding, with one point made explicit: `firwin`'s first argument is the number of *taps*, not the order. The two differ by one, and for a highpass that difference is not cosmetic.

```
def design_fir_highpass(fc, fs, numtaps):

    """Hamming-window FIR highpass. numtaps must be ODD."""

    return sig.firwin(numtaps, fc, fs=fs, pass_zero=False, window="hamming")


taps = design_fir_highpass(10.0, 500.0, 101)      # order M = 100, Type I

assert np.allclose(taps, taps[::-1])              # symmetric -> linear phase
```

Requesting an even number of taps fails outright. With `numtaps=102` SciPy raises `ValueError: A filter with an even number of coefficients must have zero response at the Nyquist frequency` — the practical face of the Type II constraint of (8). A student who reads $M = 101$ as the filter *order* and passes `M+1 = 102` therefore cannot obtain a highpass at all. This fixes the interpretation: 101 taps, order 100.

The length is also the minimum that works. By (5) the Hamming transition width is $\Delta f \approx 3.3 \times 500 / 100 =$ **16.5 Hz** — larger than the 10 Hz cutoff itself. The transition band therefore spans roughly 1.75 Hz to 18.25 Hz, placing the 2 Hz interferer barely inside the stopband. Reducing the order would widen the transition further and expose the interferer; the specification and the order are matched, not independently chosen.

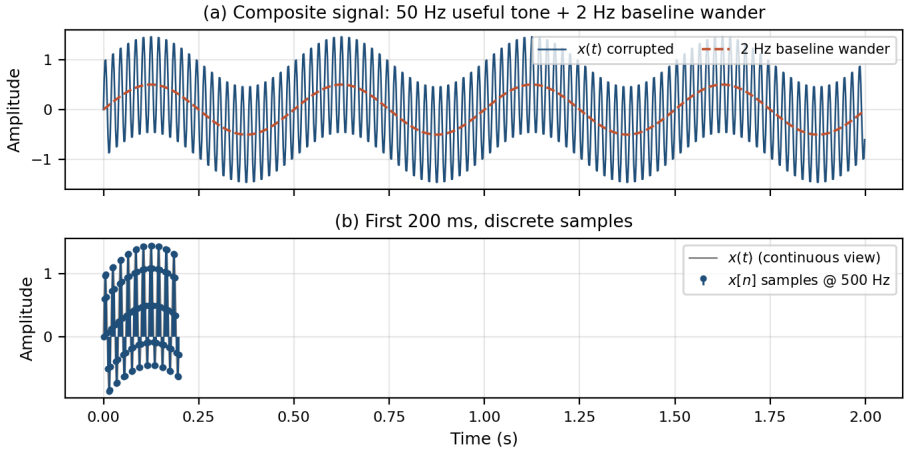


Figure 3. The corrupted test signal: the 50 Hz tone rides on the 2 Hz baseline wander.

4.2 Task 2.1 — Frequency response and verification of linear phase

```
w, h = sig.freqz(taps, worN=200_000, fs=500.0)

mag_db = 20*np.log10(np.abs(h) + 1e-300)

phase  = np.unwrap(np.angle(h))

w_gd, gd = sig.group_delay((taps, 1.0), w=8192, fs=500.0)


# linearity test: fit a straight line to the passband phase and check residual

pb = (w >= 20) & (w <= 200)

slope, icept = np.polyfit(w[pb], phase[pb], 1)

print("delay from phase slope :", -slope*500.0/(2*np.pi), "samples")

print("max deviation from line:", np.max(np.abs(phase[pb]-(slope*w[pb]+icept))))
```

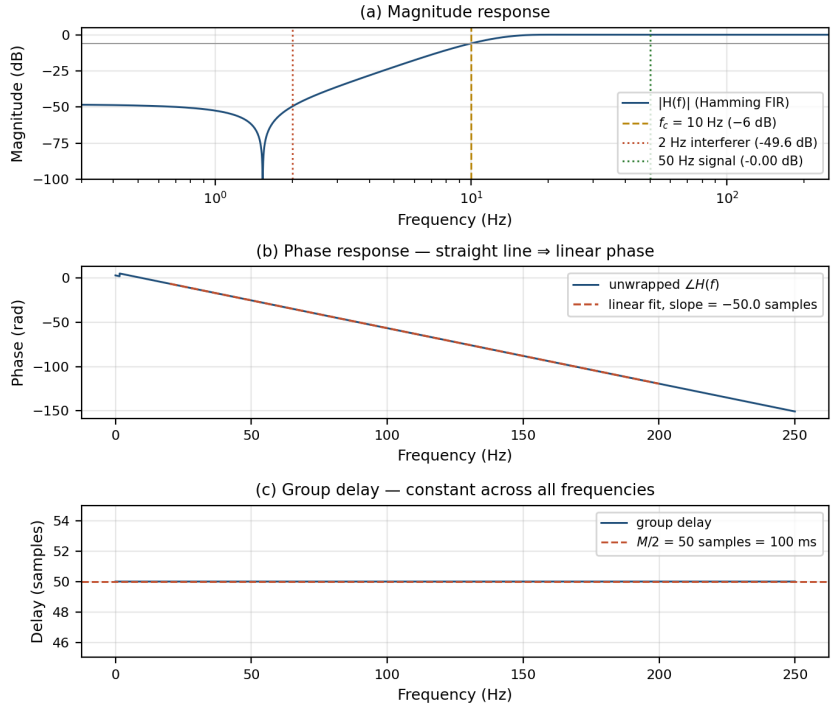


Figure 4. (a) magnitude, (b) unwrapped phase with linear fit, (c) group delay. The phase is a straight line and the delay is flat.

Measured quantity	Value	Interpretation
|H| at 2 Hz	−49.6 dB	interferer removed
|H| at 50 Hz	−0.00 dB	signal untouched

Measured quantity	Value	Interpretation
|H| at 10 Hz	−6.00 dB	firwin cutoff convention
Group delay	50.0 samples (100 ms)	= M/2 exactly
Group-delay variation	1.5×10^{-10} samples	constant to machine precision
Phase-fit residual	7.1×10^{-14} rad	phase is exactly linear
2 Hz suppression achieved	49.8 dB	0.5 → 0.0017 amplitude

Table 2. Verification of the FIR highpass specification.

The phase plot is a straight line, and fitting it returns a slope of exactly 50.00 samples with a maximum residual of 7.1×10^{-14} rad — linear to the limit of double precision. The group delay is correspondingly flat at 50 samples with a spread of 1.5×10^{-10} samples. Both confirm (7) with $M/2 = 50$. The apparent discontinuities in the raw phase plot are the $\pm\pi$ wraps at each stopband zero crossing, not phase distortion; unwrapping removes them.

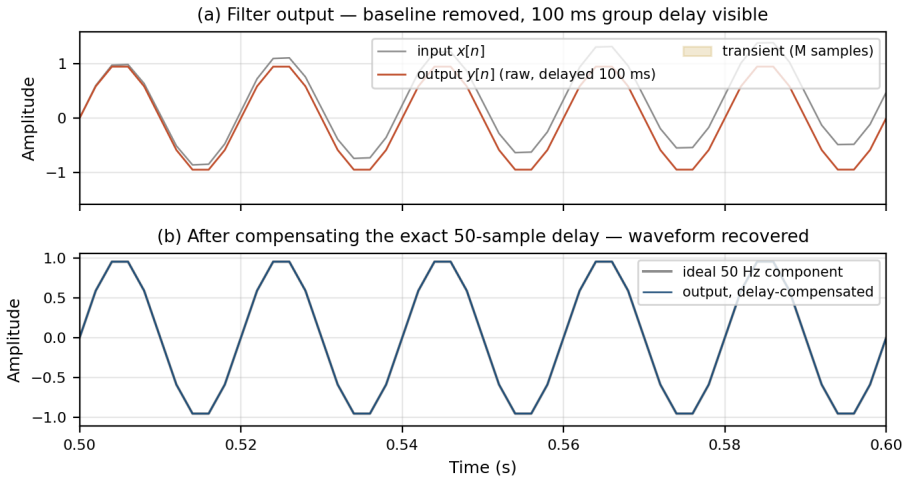


Figure 5. Filter output: baseline removed, with the constant 100 ms delay visible in (a) and exactly compensated in (b).

The output confirms the design: the 2 Hz component is reduced by 49.8 dB (amplitude 0.5 → 0.0017) while the 50 Hz component passes at unity. Two practical effects are visible. The first $M = 100$ samples are a start-up transient, during which the filter's delay line is still filling — real deployments discard or pre-charge this region. The second is the 100 ms bulk delay, which panel (b) removes with a single `np.roll(y, -50)`. That this works at all — that one shift restores the waveform exactly — *is* the practical meaning of linear phase.

4.3 Critical inference — why phase linearity matters for ECG

The importance of linear phase is easiest to see on a signal whose *shape* carries the information. A clinical ECG is read morphologically: the height of the R wave, the width of the QRS complex, and above all the level of the ST segment relative to the isoelectric baseline, since ST elevation or depression is the primary electrocardiographic evidence of myocardial ischaemia and infarction. These features are transients built from many frequency components in fixed phase relationships. A filter with non-constant group delay delays those components by different amounts, so they recombine in the wrong relative positions and the complex is reshaped — not merely shifted.

To test this, the same synthetic ECG plus baseline wander was filtered with the linear-phase FIR and with a 4th-order Butterworth IIR at the same 10 Hz cutoff. To make the comparison fair, the IIR output was given the *best possible* integer time alignment against the reference — the most generous correction a single delay can provide.

```
y_fir = np.roll(sig.lfilter(taps, 1.0, ecg_noisy), -50)   # exact compensation

b4, a4 = sig.butter(4, 10.0, btype="high", fs=500.0)

y_iir = sig.lfilter(b4, a4, ecg_noisy)                    # non-linear phase

y_zp  = sig.filtfilt(b4, a4, ecg_noisy)                   # zero-phase control

# score each against the true ECG after best-possible alignment
```

Filter	Shape correlation r	Max error (mV)
FIR, linear phase, delay-compensated	0.734	0.383
IIR Butterworth, best possible shift	0.592	0.561
Same IIR + filtfilt (zero phase)	0.696	0.390

Table 3. Waveform-shape preservation. The IIR is given optimal alignment, yet still loses.

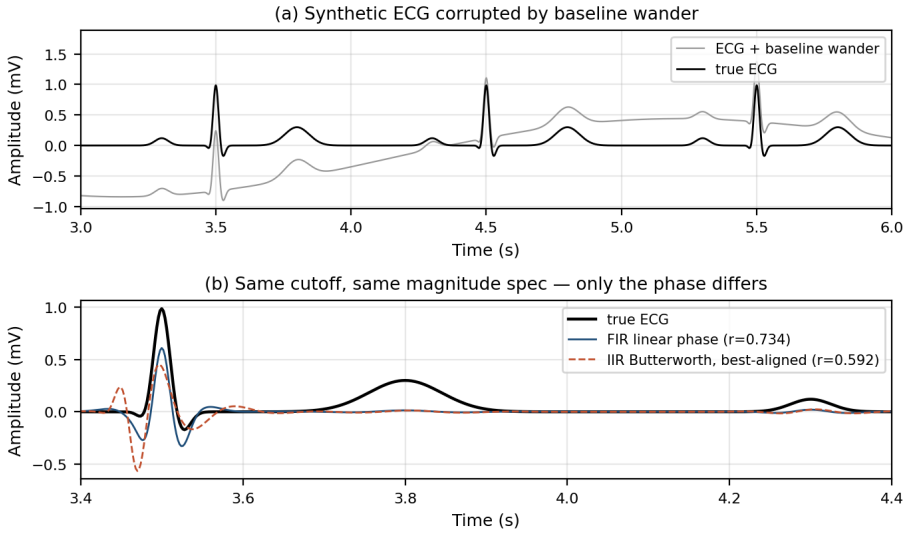


Figure 6. Synthetic ECG. Both filters have the same cutoff and comparable magnitude response — only the phase differs.

The result is decisive and, importantly, it isolates phase as the cause. The linear-phase FIR scores r = 0.734; the IIR reaches only 0.592 even with optimal alignment, because *no single time shift can undo a delay that varies with frequency*. The control experiment settles it: running the *same* IIR forward and backward with `filtfilt` cancels its phase response entirely while leaving the squared magnitude response, and the score recovers to 0.696 — close to the FIR. Since only the phase changed between the two IIR runs, the lost morphology was a phase effect, not a magnitude effect. Clinically, the reshaping seen in Figure 6 is exactly the mechanism that can manufacture apparent ST deviation in a patient who has none, which is why diagnostic ECG standards mandate linear-phase (or zero-phase) filtering. The same argument applies to image data, where non-linear phase displaces edges by different amounts depending on their spatial frequency, visibly smearing them.

One caveat should be stated plainly, and it is the natural viva question. A 10 Hz highpass is appropriate for this lab's 50 Hz test tone, but it is far too aggressive for a real diagnostic ECG, whose T wave occupies roughly 1–8 Hz; that is why all three correlations in Table 3 cap out near 0.7. Diagnostic standards call for a cutoff near 0.67 Hz or below. Realising that with the same Hamming design would require, by (5), about **2750 taps** and a group delay of 2.75 s — acceptable offline, impossible for real-time monitoring. This is the real engineering trade-off behind the order choice: linear phase at a low cutoff is expensive, and the usual escape is either an IIR with zero-phase offline filtering, or accepting a higher cutoff in monitoring mode.

4.4 Task 2.2 — The quantizer

The quantizer implements (10) directly. The clipping bound is asymmetric, $[-V_{\max}, V_{\max} - \Delta]$, because a b-bit two's-complement code has one more negative level than positive — a detail that matters in §4.5.

```
def uniform_mid_tread_quantizer(x, bits, V_max):

    """Uniform mid-tread quantizer with saturation.

    L = 2**bits levels, step delta = 2*V_max/L, zero IS a level."""

    L = 2**bits

    delta = (2.0*V_max) / L

    xq = np.round(x/delta) * delta            # mid-tread rounding

    return np.clip(xq, -V_max, V_max - delta) # saturation (asymmetric)
```

```
e = x - xq                                  # quantization error, Eq. (11)
sqnr = 10*np.log10(np.mean(x**2) / np.mean(e**2))
```

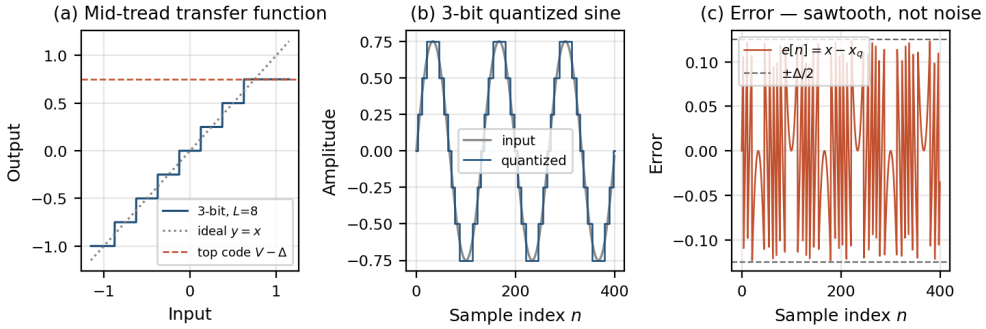


Figure 7. (a) the mid-tread staircase, (b) a 3-bit quantized sine, (c) the error — a deterministic sawtooth bounded by ±Δ/2, not random noise.

Panel (c) already contains the central insight of the whole task. The error is not noise: it is a smooth, repeating sawtooth locked to the signal, whose period is set by how fast the signal traverses the quantization levels. It *resembles* noise only when the levels are fine enough and the signal busy enough that successive errors are effectively unpredictable.

4.5 Task 2.2 — SQNR and the 6 dB-per-bit rule

SQNR was measured for $b = 2 \ldots 16$ using a coherently sampled sine (93 whole cycles in 8192 samples, so no spectral leakage). Two amplitudes were used: $A = V_{max}$ as the naive full-scale choice, and $A = V_{max} - \Delta$, which is the largest amplitude that provably never saturates.

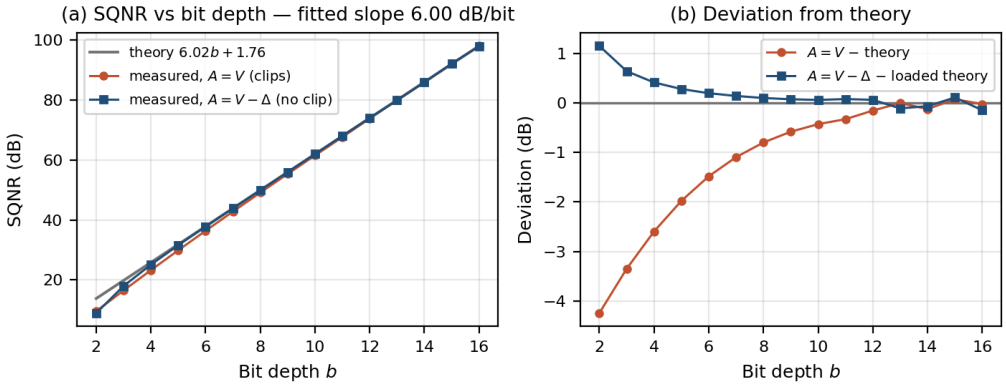


Figure 8. (a) measured versus theoretical SQNR; (b) deviation from theory, separating clipping from model breakdown.

b	SQNR, A = V (dB)	Dev.	Clipped	SQNR, A = V−Δ (dB)	Dev. vs Eq.(14)
2	9.56	−4.24	1885	8.93	+1.15
3	16.47	−3.35	1317	17.96	+0.64
4	23.24	−2.60	927	25.09	+0.41
8	49.12	−0.80	231	49.95	+0.10
12	73.84	−0.16	57	74.06	+0.06
16	98.06	−0.02	15	97.93	−0.15

Table 4. SQNR against theory. Fitting the no-clip data for $b \geq 8$ gives a slope of 6.005 dB/bit.

The rule itself is confirmed cleanly: a least-squares fit over the well-behaved region returns **6.005 dB/bit** against the predicted 6.02, and at $b = 16$ the measured 98.06 dB matches the theoretical 98.08 dB to within 0.02 dB. The interesting content is in the deviations, which have *two distinct causes* that Table 4 separates.

Cause 1 — saturation, and it dominates. With $A = V_{max}$, the sine peaks at exactly V_{max}, but the highest available code is $V_{max} - \Delta$. Every sample near a positive peak is therefore clipped, producing errors of up to Δ — twice the $\Delta/2$ that (12) assumes. This alone accounts for −4.24 dB at $b = 2$. It shrinks with bit depth (1885 clipped samples at $b = 2$, only 15 at $b = 16$) because the fraction of a sine period spent within Δ of the peak falls as Δ does. Backing the amplitude off by a single LSB removes the effect entirely: zero clipped samples at every bit depth.

Cause 2 — the noise model itself. Once clipping is removed and the loading factor of (14) is applied, a residual remains that is largest exactly where the model should be weakest: +1.15 dB at $b = 2$, falling to +0.06 dB at $b = 12$. Figure 9 diagnoses it directly.

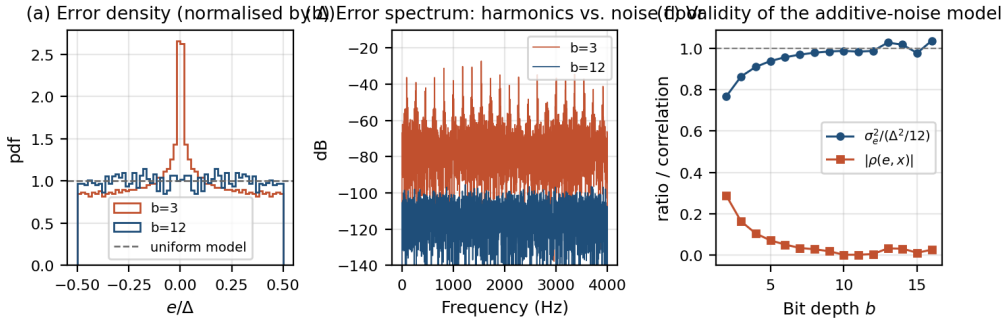


Figure 9. (a) error distribution against the uniform model; (b) error spectrum; (c) validity of the additive-noise assumptions versus bit depth.

b	σ² / (Δ²/12)	|corr(e, x)|	THD (%)
2	0.768	0.287	—
3	0.864	0.163	11.21
4	0.911	0.104	4.28
8	0.979	0.028	0.026
12	0.988	0.005	0.0024
16	1.038	0.026	0.00015

Table 5. The additive-white-noise model degrades systematically as bit depth falls.

Every assumption behind (12) fails at low b, and each failure is measurable. The error is *not independent of the signal*: its correlation with x rises from 0.005 at $b = 12$ to 0.287 at $b = 2$. It is *not uniformly distributed*: its variance is only $0.768 \times \Delta^2/12$ at $b = 2$, because a sine does not sweep the quantizer levels uniformly — it lingers near its peaks — so the true variance is smaller than the model assumes, which is why the measured SQNR comes out slightly *better* than the corrected theory rather than worse. And it is *not white*: Figure 9(b) shows the $b = 3$ error concentrated in discrete harmonics of the input rather than spread as a floor, giving a total harmonic distortion of 11.2%, against 0.0024% at $b = 12$. At low bit depth, quantization error is audible and visible as *distortion*, not as noise — which is precisely the problem Task 2.3 addresses.

4.6 Task 2.3 (extension) — Dithering

A 4-bit quantizer was applied to a steady tone (chosen coherent with the analysis FFT so that any energy at harmonic frequencies is unambiguously created by the quantizer), first directly and then with dither added before quantization, per (15).

```
delta = 2*V_max / 2**4

q_plain = uniform_mid_tread_quantizer(audio, 4, V_max)


# dither is added BEFORE the quantizer -- adding it after does nothing

gauss = rng.normal(0, delta/2, audio.size)

q_gauss = uniform_mid_tread_quantizer(audio + gauss, 4, V_max)
```

```
tpdf = (rng.random(audio.size) - rng.random(audio.size)) * delta   # +-1 LSB
q_tpdf = uniform_mid_tread_quantizer(audio + tpdf, 4, V_max)
```

Condition	Noise power	SQNR (dB)	THD (%)	|corr(e,x)|
No dither	1.19×10^{-3}	25.09	3.553	0.106
Gaussian, $\sigma = \Delta/2$	4.99×10^{-3}	18.85	1.102	0.041
TPDF, ±1 LSB	3.69×10^{-3}	20.16	0.872	0.015

Table 6. Dithering trades measured SQNR for a large reduction in harmonic distortion.

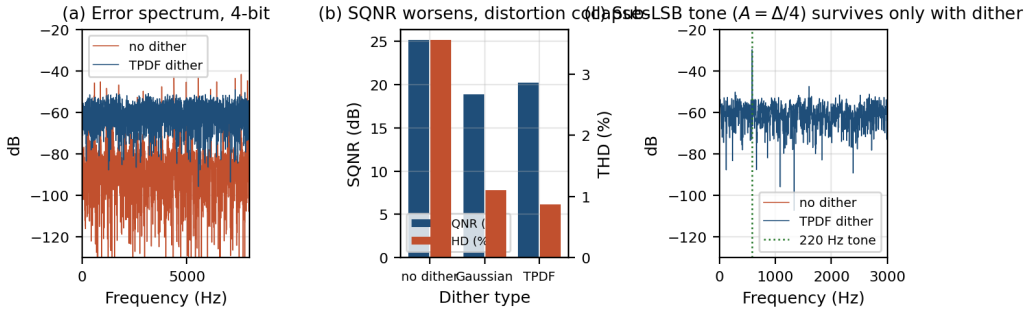


Figure 10. (a) error spectrum with and without dither; (b) the SQNR/THD trade; (c) a tone of amplitude Δ/4 survives only with dither.

The paradox in the task title is now quantitative. Dither makes the signal-to-noise ratio *measurably worse* — TPDF costs 4.93 dB, Gaussian 6.24 dB — while cutting harmonic distortion by a factor of four and the error–signal correlation by a factor of seven. The resolution is that SQNR treats all error energy as equivalent, and the ear does not. Undithered, the error sits in discrete harmonics of the signal: these are locked to the music, track it in pitch, and are perceived as a harsh metallic edge on the sound itself. Dithered, the same energy (plus a little more) is spread into a constant, signal-independent broadband hiss, which the auditory system readily separates from the signal and largely ignores. Trading a small amount of benign noise for a large amount of correlated distortion is a good bargain perceptually, even though it is a bad one numerically. TPDF outperforms Gaussian on both axes at once, which is why it is the standard choice.

Panel (c) makes the deeper point. A tone of amplitude Δ/4 — a quarter of one quantization step — was quantized at 4 bits. Without dither the output contains exactly *one* distinct level: the signal never crosses a decision threshold, the output is identically zero, and the tone is annihilated with output power of exactly 0.0. With TPDF dither the output takes three levels and the 220 Hz tone reappears clearly above the noise floor. Dither does not merely mask distortion; it *linearises* the quantizer, letting it convey information about signals smaller than its own resolution by encoding them in the statistics of the output rather than in any single sample. This is the same principle that underlies oversampling converters, and it is why professional audio is dithered whenever bit depth is reduced.

5. Conclusion

A 101-tap Hamming FIR highpass was designed and verified against its specification: 49.8 dB suppression of the 2 Hz baseline wander, −0.00 dB on the 50 Hz signal, and phase linear to 7.1×10^{-14} rad with a group delay constant at exactly $M/2 = 50$ samples. Two design constraints that the brief leaves implicit were established: a highpass FIR requires an odd number of taps, since a Type II filter forces a zero at Nyquist and SciPy refuses the design; and 101 taps is the minimum length that realises a 10 Hz cutoff, because the Hamming transition width of 16.5 Hz already exceeds the cutoff. The worth of linear phase was demonstrated on a synthetic ECG, where the FIR preserved waveform morphology (r = 0.734) against 0.592 for a Butterworth IIR granted optimal time alignment — and the zero-phase `filtfilt` control, scoring 0.696 with the same IIR, isolated phase rather than magnitude as the cause. On the quantization side, a mid-tread quantizer reproduced the 6 dB/bit rule to a fitted 6.005 dB/bit and matched theory within 0.02 dB at 16 bits. The deviations at low bit depth were decomposed into saturation against the asymmetric top code, which dominates and disappears when the amplitude is backed off by one LSB, and genuine failure of the additive-white-noise model, quantified by an error–signal correlation rising to 0.287, an error variance falling to $0.768 \times \Delta^2/12$, and a total harmonic distortion of 11.2% at 3 bits. Dithering was then shown to trade 4.9 dB of SQNR for a fourfold reduction in that distortion,

and — most tellingly — to let a 4-bit quantizer reproduce a tone of amplitude $\Delta/4$ that was otherwise annihilated entirely. Taken together, the two halves of the lab describe the same engineering principle from opposite ends: the filter's linear phase and the quantizer's dither are both deliberate costs accepted in one metric to protect a property that matters more in practice.

LAB 3:

Multirate DSP and Adaptive Filtering

Decimation and Interpolation with Anti-Aliasing and Anti-Imaging Filters, LMS Adaptive Cancellation of Powerline Interference, and Fractional Sampling-Rate Conversion

Summary

This report covers two ways of changing a signal without changing what it means: altering the rate at which it is sampled, and removing interference whose parameters are not known in advance. A two-tone sequence (1 kHz at amplitude 1.0 and 4 kHz at amplitude 0.7, sampled at 10 kHz) is decimated by $D = 2$ and interpolated by $I = 3$. The decimation result is unusually instructive: the 4 kHz component folds to exactly 1 kHz and arrives in *anti-phase* with the genuine 1 kHz tone, so naive decimation does not add a spurious line — it *subtracts* from the wanted one, reducing its amplitude from 1.000 to 0.300 = $A_1 - A_2$, verified to 1.0×10^{-12}. With equal amplitudes the signal is annihilated completely (residual 1.5×10^{-12}). Inserting an anti-aliasing lowpass before the down-sampler restores the tone to 0.9997. Interpolation shows the dual failure — spectral images at 6, 9, 11 and 14 kHz — removed to −76.3 dB by an anti-imaging filter which must carry a passband gain of I to undo the amplitude loss of zero-stuffing (0.539 → 1.617). An LMS adaptive noise canceller was then written from scratch and compared against a static IIR notch on a 60 Hz-corrupted ECG. When the mains frequency is exactly known, the notch wins (52.5 dB against 44.6 dB); when it drifts by only 0.8 Hz, the notch collapses to 4.2 dB while the LMS is unaffected at 44.4 dB. The autocorrelation matrix of a sinusoidal reference is shown to have rank 2 regardless of filter length, which explains both why $M = 2$ taps suffice in principle and why the practical bound $\mu < 2/\mathrm{tr}(R) = 1.0$ predicts the observed divergence far better than the classical $\mu < 2/\lambda_{\max} = 1.91$. Finally, 44.1 kHz audio is converted to 48 kHz through a single combined filter at $I/D = 160/147$, preserving all test tones to within 0.27% while a polyphase implementation reduces the arithmetic by a factor of 23,520.

1. Objective and System View

The objective is to establish three results. First, that changing the sampling rate is never a matter of simply adding or removing samples: down-sampling folds out-of-band content into the baseband, up-sampling creates spectral images, and each requires a filter placed on the correct side of the rate change — before the down-sampler, after the up-sampler. Second, that a filter designed around a fixed assumption is only as good as that assumption, whereas an adaptive filter *learns* its own coefficients from the data and therefore survives when the assumption fails. Third, that these two ideas meet in practical rate conversion, where a single filter can serve both roles and a polyphase decomposition makes an otherwise impossible computation routine.

The tasks are sequenced so that each supplies the motivation for the next. Task 3.1 demonstrates what the correct filter buys in a fully deterministic setting where the answer can be computed by hand. Task 3.2 moves to a setting where the interference parameters are *not* known exactly, which is precisely where a fixed design fails and adaptation is required. Task 3.3 returns to multirate with a realistic ratio, where the naive implementation is computationally absurd and the structural insight matters more than the filter design.

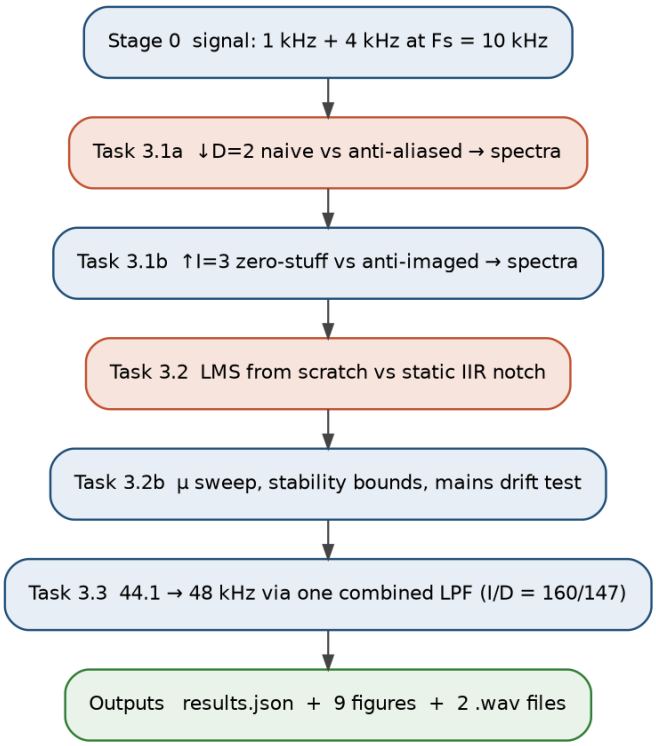


Figure 1. Experiment flowchart. Each task motivates the next.

2. Mathematical (Theoretical) Foundation

2.1 Decimation and the folding rule

Down-sampling by an integer D keeps every D-th sample,

$$y[m] = x[mD], \quad \text{new rate } F_s' = F_s/D \qquad (1)$$

In the frequency domain this stretches and superimposes D shifted copies of the spectrum:

$$Y(e^{j\omega}) = (1/D) \sum_{k=0}^{D-1} X\left(e^{j(\omega - 2\pi k)/D} \right) \qquad (2)$$

The copies overlap unless the signal is bandlimited to the new Nyquist frequency $F_s/2D$. Any component above it reappears at the folded frequency

$$f_{\text{apparent}} = | f - F_s' \cdot \text{round}(f / F_s') | \qquad (3)$$

This is why the anti-aliasing filter must run **before** the down-sampler: once samples are discarded, the folded copies are numerically superimposed and no later filter can separate them. Its cutoff must satisfy

$$f_c < F_s / (2D) \qquad (4)$$

A point usually left implicit is that folding is *coherent*: the alias carries the phase of its parent component, so it *adds as a phasor* to whatever already occupies the destination frequency. For the signal used here, $\sin(2\pi \cdot 4000 \cdot n/10000)$ sampled at 5 kHz becomes $\sin(2\pi \cdot 0.8n) = -\sin(2\pi \cdot 0.2n)$, so the alias arrives inverted and the two components *subtract*:

$$y[m] = (A_1 - A_2) \cdot \sin(2\pi \cdot 1000 \cdot m/5000) \qquad (5)$$

2.2 Interpolation and spectral imaging

Up-sampling by I inserts $I-1$ zeros between samples:

$$y[m] = x[m/I] \text{ if } m \text{ is a multiple of } I, \text{ else } 0 \qquad (6)$$

Zero-stuffing does not change the DTFT values at all; it merely reinterprets them on a faster grid,

$$Y(e^{j\omega}) = X(e^{j\omega I}) \qquad (7)$$

so the spectrum, previously periodic with period F_s, is now periodic with period F_s inside a band that extends to $IF_s/2$. The result is $I-1$ unwanted *images* of the baseband. An anti-imaging lowpass placed **after** the up-sampler removes them, with cutoff

$$f_c < IF_s / (2I) = F_s/2 \qquad (8)$$

One consequence is easily missed. Zero-stuffing leaves only one non-zero sample in every I, so the mean power falls; after a unity-gain lowpass the reconstructed waveform has amplitude A/I. The anti-imaging filter must therefore carry a passband gain of I:

$$h_{\text{anti-image}}[n] = I \cdot h_{\text{LPF}}[n] \qquad (9)$$

2.3 Rational rate conversion

A non-integer ratio is realised by cascading the two operations,

$$F_{\text{out}} = F_{\text{in}} \cdot I / D, \quad \text{with } I/D \text{ in lowest terms} \qquad (10)$$

with the up-sampler first, so that no information is discarded before it can be used. Between them sit two lowpass filters — anti-imaging then anti-aliasing — running at the same intermediate rate, so they can be merged into *one* filter whose cutoff is the more restrictive of the two:

$$f_c = \min(F_{\text{in}}/2, F_{\text{out}}/2) \qquad (11)$$

Implemented directly this is hopeless: for 44.1 → 48 kHz the intermediate rate is $44100 \times 160 = 7.056$ MHz. The polyphase insight is that $I-1$ of every I filter inputs are zero and only one output in D is kept, so each output sample requires only N/I multiplications instead of ND, a saving of

$$\text{speed-up} = ND / (N/I) = I \cdot D \qquad (12)$$

2.4 The LMS adaptive noise canceller

An adaptive noise canceller receives a *primary* input $d[n] = s[n] + v[n]$ (signal plus interference) and a *reference* input $x[n]$ that is correlated with the interference v but not with the signal s — for powerline cancellation, a direct measurement of the mains. An FIR filter h forms an estimate of the interference and it is subtracted:

$$y(n) = h^{\mathrm{T}}(n)\, x(n), \qquad e(n) = d(n) - y(n) \qquad (13)$$

Because s and x are uncorrelated, minimising $E[e^2]$ with respect to h cannot remove any of s — it can only cancel the part of d that the reference can explain. Hence the *error signal is the cleaned output*, which is the conceptual heart of the method. The Wiener solution is $h^* = R^{-1}p$, with $R = E[xx^{\mathrm{T}}]$ and $p = E[dx]$. LMS replaces the true gradient by its instantaneous estimate, giving the stochastic-gradient update

$$h(n+1) = h(n) + \mu \cdot e(n) \cdot x(n) \qquad (14)$$

Convergence of the mean weight vector requires every mode to decay, i.e. $|1 - \mu\lambda_i| < 1$ for all eigenvalues of R, giving

$$0 < \mu < 2 / \lambda_{\max} \qquad (15)$$

Convergence of the mean *square* error is stricter and is the bound that matters in practice:

$$0 < \mu < 2 / \operatorname{tr}(R) = 2 / (M \cdot P_x) \qquad (16)$$

Since $\operatorname{tr}(R) = \sum\lambda_i \geq \lambda_{\max}$, (16) is always the tighter condition. The price of a larger step size is a larger residual error even after convergence, quantified by the misadjustment

$$\mathcal{M} \approx \mu \cdot \operatorname{tr}(R) / 2 \qquad (17)$$

so convergence speed and steady-state accuracy are traded directly against one another through the single parameter μ.

3. System Block Diagram

The three chains are shown below. The two multirate chains differ only in the *order* of the filter and the rate changer — and that order is the whole content of Task 3.1. In the canceller, note that the adaptation path is driven by the output $e[n]$: the filter is steered by its own error, which is what makes the structure self-tuning.

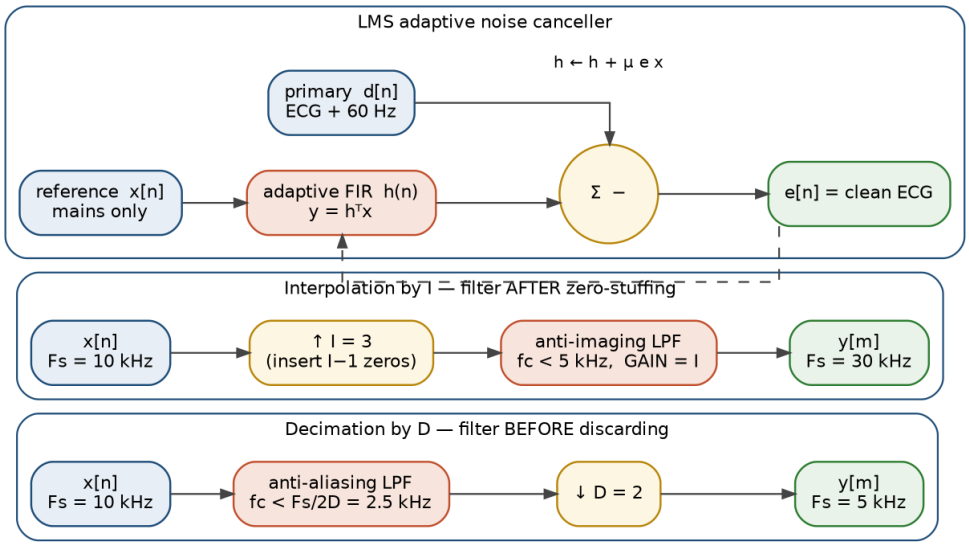


Figure 2. Decimation, interpolation, and the LMS adaptive noise canceller.

4. Implementation and Results

All results come from one reproducible NumPy/SciPy script. Record lengths were chosen coherent with the analysis FFT at every rate, so each tone occupies exactly one bin and the measured amplitudes are exact rather than leakage-limited. Table 1 lists the parameters.

Parameter	Value	Notes
Input rate F_s	10 kHz	Nyquist 5 kHz
Tones	1 kHz (A=1.0), 4 kHz (A=0.7)	both initially valid
Decimation D	2	new Nyquist 2.5 kHz
Anti-aliasing cutoff	2.2 kHz (limit 2.5 kHz)	Eq. (4), 201 taps
Interpolation I	3	new rate 30 kHz
Anti-imaging cutoff	4.5 kHz, gain = 3	Eqs. (8)–(9), 301 taps
ECG rate / mains	500 Hz / 60 Hz (A = 0.5)	SNR before −7.17 dB
LMS	M = 4, μ = 0.005	written from scratch
Resampling	44.1 → 48 kHz, I/D = 160/147	3201-tap combined LPF

Table 1. Experiment parameters.

4.1 Task 3.1 — Decimation by D = 2

Before down-sampling, (3) predicts where each component will land at the new 5 kHz rate: the 1 kHz tone is safely below the new 2.5 kHz Nyquist, while the 4 kHz tone folds to |4000 − 5000| = **1000 Hz** — precisely on top of the tone we want to keep.

```
fold = lambda f, fs: abs(f - fs*round(f/fs))

print(fold(4000, 5000))          # -> 1000.0 Hz : lands on the 1 kHz tone


x_naive = x[::D]                              # discard every 2nd sample
```

```
# correct route: band-limit FIRST, then discard
h_aa   = sig.firwin(201, 2200, fs=10000, window="hamming")   # fc < Fs/2D
x_dec  = sig.lfilter(h_aa, 1.0, x)[::D]
```

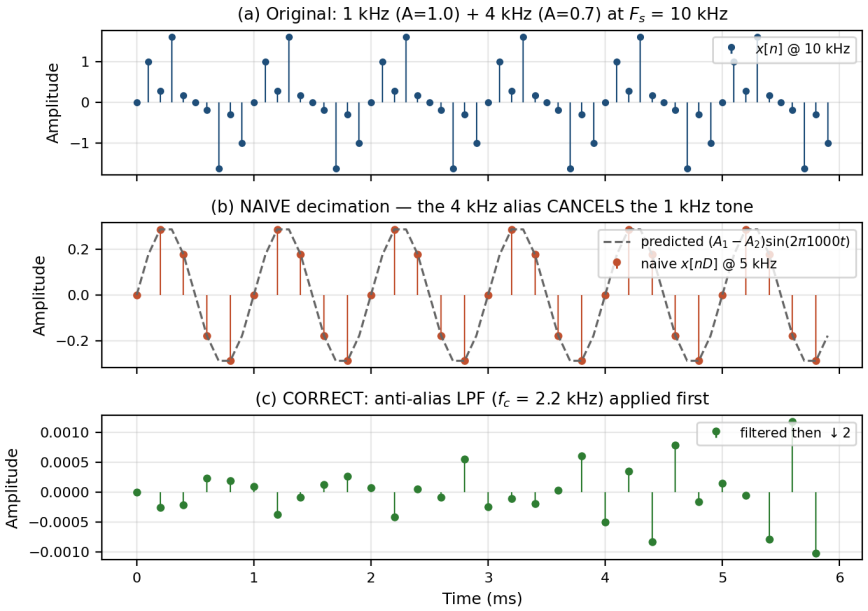


Figure 3. Decimation in the time domain. Naive down-sampling does not add a tone — it cancels the one we wanted.

Component	Original	Naive ↓ 2	LPF then ↓ 2
1 kHz amplitude	1.000	0.300	0.9997
4 kHz amplitude	0.700	folded to 1 kHz	removed
Prediction ($A_1 - A_2$)	—	0.300 ✓	—
Match to Eq. (5)	—	1.0×10^{-12}	—

Table 2. Measured amplitudes. Aliasing here is destructive, not additive.

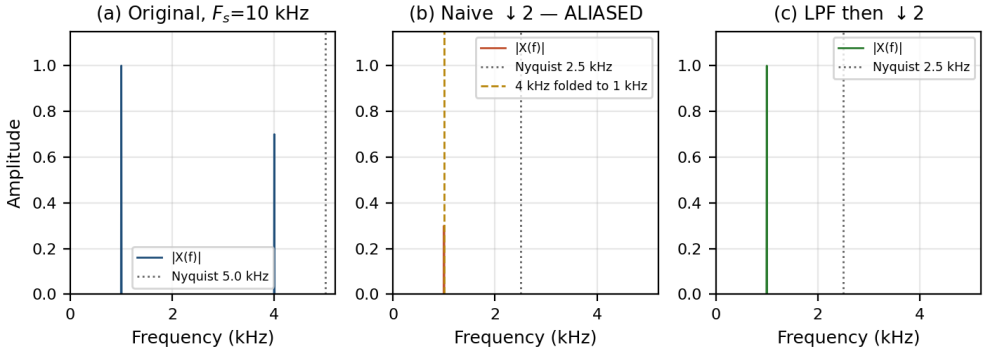


Figure 4. Spectra: (a) both tones resolved; (b) naive decimation leaves a single, weakened 1 kHz line; (c) filtering first preserves it.

The result is more interesting than the textbook description of aliasing suggests. The usual account is that a spurious line *appears* at the folded frequency. Here the folded line lands on an existing one, and because folding preserves phase, the two combine as phasors. For a sine, sampling at $F_s' = 5$ kHz gives $\sin(2\pi \cdot 0.8m) = -\sin(2\pi \cdot 0.2m)$, so the alias arrives **inverted** and subtracts. The measured 1 kHz amplitude after naive decimation is 0.300 — exactly $A_1 - A_2 = 1.0 - 0.7$ — and the whole decimated sequence matches (5) to 1.0×10^{-12}. A 30% error in the signal we were trying to preserve has been introduced by an operation that never touched it directly.

The limiting case is worth stating because it is so counter-intuitive. With $A_1 = A_2$, the cancellation is total: the decimated sequence is zero to a residual of 1.5×10^{-12} — a perfectly good two-tone signal, sampled correctly at 10 kHz, is reduced to silence by discarding every second sample. This is the strongest possible argument for the anti-aliasing filter, and it

also disposes of a common misconception: aliasing damage cannot be detected by looking for unexpected energy in the output, because the damage may consist of energy that has *disappeared*. With the filter in place the tone is preserved at 0.9997.

4.2 Task 3.1 — Interpolation by I = 3

```
x_zs = np.zeros(len(x)*I);  x_zs[::I] = x        # zero-stuffing, Eq. (6)


# anti-imaging LPF: cutoff < I*Fs/(2I) = 5 kHz, and PASSBAND GAIN = I
h_ai  = I * sig.firwin(301, 4500, fs=10000*I, window="hamming")
x_int = sig.lfilter(h_ai, 1.0, x_zs)
```

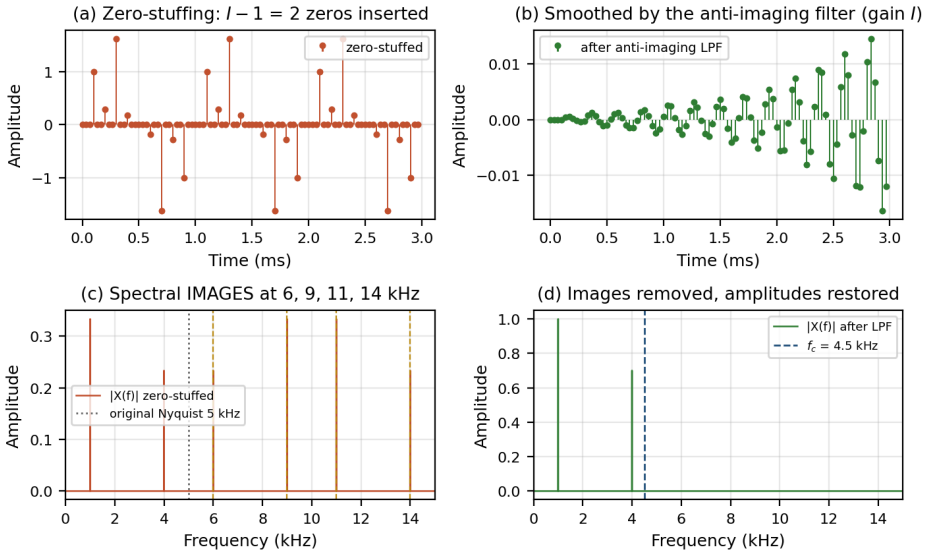


Figure 5. Interpolation: (a) zero-stuffed sequence, (b) smoothed output, (c) images at 6/9/11/14 kHz, (d) images removed.

Zero-stuffing is the exact dual of decimation. It does not corrupt anything — by (7) the original spectral values survive untouched — but it makes the spectrum's inherent periodicity visible inside the new, wider Nyquist band. With F_s = 10 kHz and I = 3, the baseband tones at 1 and 4 kHz are joined by images at 6, 9, 11 and 14 kHz, exactly as predicted by reflecting each tone about multiples of 10 kHz. The time-domain view in panel (a) shows why: two out of every three samples are zero, which is a violently discontinuous waveform, and those discontinuities are the images. The anti-imaging filter suppresses them to 1.53×10^{-4} (−76.3 dB) and, in doing so, smooths the waveform back into a recognisable sinusoid.

The gain factor in (9) is the detail most easily overlooked, and it is not cosmetic. With a unity-gain lowpass the reconstructed peak is 0.539 against an original of 1.617 — a factor of very nearly I = 3 — because two-thirds of the samples contributed nothing to the average. Scaling the filter by I restores 1.6168, matching the input to four figures, and the 1 kHz line measures exactly 1.0000. An interpolator that omits this factor is not wrong in shape, only in level, which makes the error easy to miss until it is cascaded.

4.3 Task 3.2 — LMS adaptive noise cancellation

The canceller was implemented directly from (13)–(14), with no adaptive-filter library. The reference input is a measurement of the mains itself — physically available from a wall transformer — and is correlated with the interference but not with the ECG, which is the condition that makes the method safe.

```
def lms_adaptive_filter(noisy, reference, M, mu):
    """h(n+1) = h(n) + mu*e(n)*x(n); the ERROR is the cleaned signal."""
    h, clean = np.zeros(M), np.zeros(len(noisy))
    for k in range(M, len(noisy)):
        x_vec = reference[k:k-M:-1]     # last M samples, time-reversed
        y_hat = np.dot(h, x_vec)        # estimated interference
        e     = noisy[k] - y_hat        # error == cleaned output
```

```
        clean[k] = e
        h = h + mu * e * x_vec          # stochastic-gradient update
    return clean, h
```

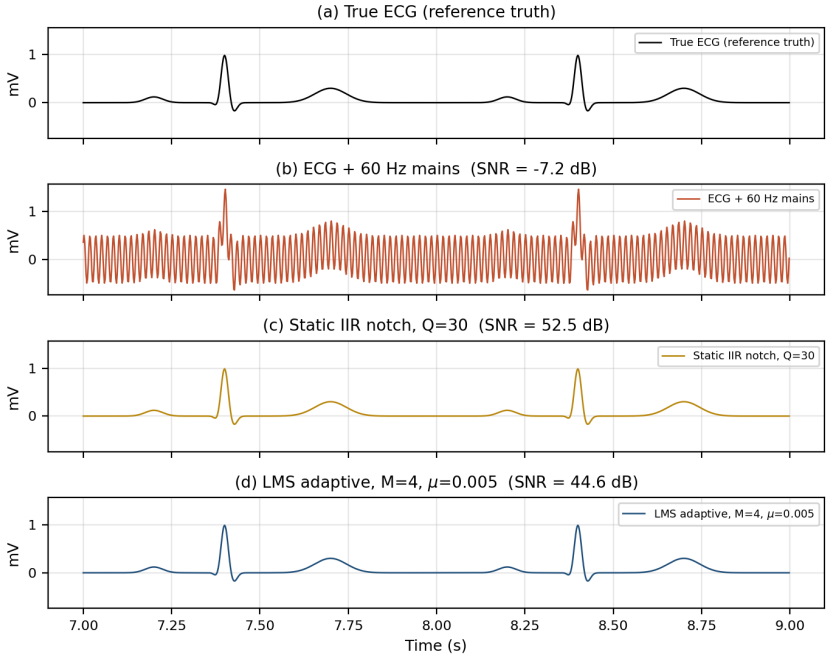


Figure 6. Time domain: true ECG, corrupted ECG, static notch output, and LMS output at μ = 0.005.

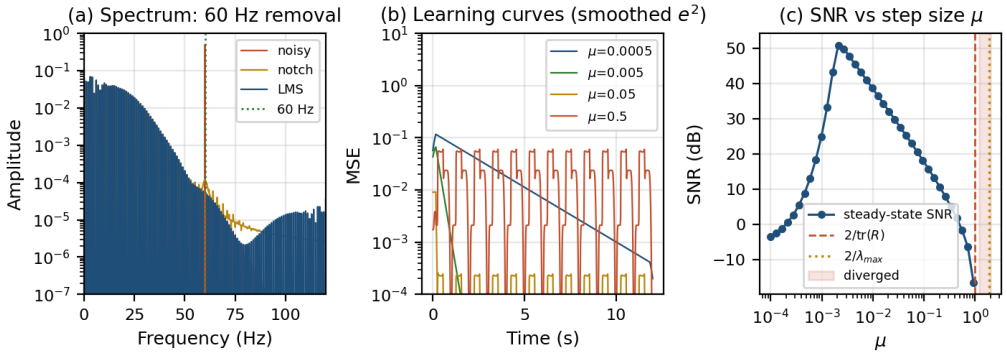


Figure 7. (a) spectra showing 60 Hz removal; (b) learning curves for four step sizes; (c) steady-state SNR against μ with the theoretical bounds marked.

Both filters remove the interference when it behaves as designed: from an input SNR of −7.17 dB, the notch reaches 52.5 dB and the LMS 44.6 dB. The learning curves in Figure 7(b) show the mechanism that distinguishes them. The LMS begins knowing nothing — all weights zero — and converges within roughly a second at $\mu = 0.005$. Smaller steps converge more slowly but settle lower; larger steps converge almost instantly but plateau at a visibly higher error floor, exactly as the misadjustment relation (17) predicts.

4.4 Task 3.2 — Step size, stability, and why divergence occurs

The reference is a pure sinusoid, and this gives the eigenvalue analysis an unusually clean structure. Its autocorrelation matrix $R[i,j] = (A^2/2)\cos(\omega(i-j))$ is Toeplitz but **rank 2** for any filter length — a single real sinusoid spans only a two-dimensional subspace, corresponding to its in-phase and quadrature components.

Quantity	M = 4 (used)	M = 8	M = 16
rank(R)	2	2	2
λ_max	1.046	2.091	4.176
tr(R) = M·P_x	2.00	4.00	8.00
2/λ_max (mean)	1.913	0.957	0.479

Quantity	M = 4 (used)	M = 8	M = 16
2/tr(R) (mean square)	1.000	0.500	0.250

Table 3. Eigenstructure of a sinusoidal reference. Rank is 2 regardless of M, but the bounds tighten as M grows.

Three consequences follow, and together they answer the question of what μ may be. First, since only two eigenvalues are non-zero, **two taps are sufficient in principle** to cancel a single tone: two weights provide exactly the two degrees of freedom — amplitude and phase — needed to synthesise any sinusoid at that frequency. Additional taps add no capability for a pure tone; they merely enlarge tr(R) and so tighten the permissible step size, which is why $M = 4$ was chosen rather than something larger.

Second, the two bounds differ by a factor of nearly two, and the experiment discriminates between them. Sweeping μ over four decades gives a clear optimum at $\mu = 0.0021$ (51.0 dB), degradation on both sides, and outright numerical divergence first observed at $\mu = 1.20$. The classical mean bound predicts stability up to $2/\lambda_{max} = 1.913$, which is optimistic — performance has already collapsed to −40 dB well before that. The mean-square bound $2/\mathrm{tr}(R) = 1.000$ is the far better predictor: at exactly $\mu = 1.0$ the output is worthless, and divergence to infinity follows immediately above it. This is the practical justification for using (16) rather than (15) when choosing a step size.

Third, the shape of Figure 7(c) is the convergence–stability trade-off made visible. At very small μ the filter is stable but has not finished converging within the record, so the measured SNR is poor for a reason unrelated to accuracy. At large μ each update over-corrects; the weight vector rattles around the optimum instead of settling, and by (17) the residual grows in direct proportion to μ. Beyond $2/\mathrm{tr}(R)$ the over-correction exceeds the error it was meant to remove, each update amplifies the last, and the weights grow without bound. The useful range is therefore bounded below by patience and above by arithmetic.

4.5 Critical inference — static notch versus adaptive filter

The comparison that matters is not how the two filters behave under ideal conditions but how they fail. The experiment was repeated with the mains frequency drifting slowly from 60.0 to 60.8 Hz — a realistic amount for a loaded grid, and less than 1.4% of the nominal value.

Condition	Unfiltered	Static notch (Q = 30)	LMS (M=4)
Mains exactly 60.0 Hz	−7.17 dB	52.54 dB	44.65 dB
Mains drifting 60 → 60.8 Hz	−7.17 dB	4.23 dB	44.41 dB
Degradation	—	−48.3 dB	−0.24 dB

Table 4. Output SNR. The notch is better when its assumption holds and useless when it does not.

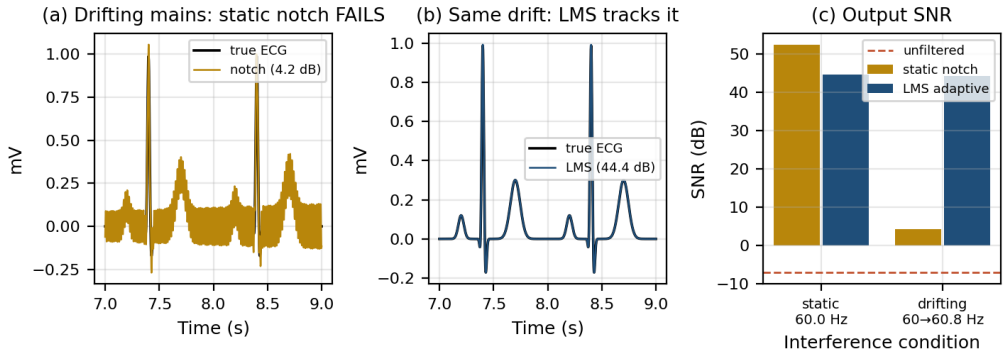


Figure 8. Under a 0.8 Hz drift the static notch fails completely while the LMS output is essentially unchanged.

The honest result is that the static notch *wins* when the interference is exactly what it was designed for — 52.5 dB against 44.6 dB — because it applies infinite attenuation at precisely the right frequency with no learning transient and no misadjustment. An adaptive filter should not be presented as universally superior. What changes the verdict is the drift: the notch loses 48.3 dB, falling to 4.2 dB, while the LMS loses 0.24 dB. The interference has simply walked out of the notch's stopband, and the filter has no mechanism to notice or respond. The LMS, by contrast, never assumed a frequency at all; it tracks whatever the reference presents.

The trade-off is worse than a single number suggests, because the obvious fix makes it worse. A notch also distorts the signal it is protecting, since the ECG has genuine spectral content at 60 Hz, and that distortion is reduced by narrowing the notch. But narrowing it also reduces the tolerance to drift.

Q	−3 dB width	ECG distortion	SNR under drift
10	6.00 Hz	58.73 dB	20.10 dB
30	2.00 Hz	69.50 dB	4.23 dB
100	0.60 Hz	72.18 dB	−5.03 dB
200	0.30 Hz	72.27 dB	−6.55 dB

Table 5. The notch designer's dilemma: reducing distortion by raising Q destroys drift tolerance.

Raising Q from 10 to 200 improves ECG fidelity from 58.7 to 72.3 dB, but drift performance collapses from 20.1 dB to −6.6 dB — an output *worse than doing nothing at all*, because the notch now removes ECG content while leaving the interference untouched. There is no value of Q that is simultaneously gentle on the signal and tolerant of drift; the two requirements are in direct opposition, and the parameter that trades between them must be fixed at design time when the drift is not yet known. That is the precise sense in which the problem is not solvable by a static filter, and it is the reason adaptive filtering exists.

4.6 Task 3.3 (extension) — Fractional resampling, 44.1 kHz to 48 kHz

The ratio is fixed by arithmetic: gcd(48000, 44100) = 300, so I/D = 160/147 in lowest terms. Reducing the fraction matters — an unreduced 48000/44100 would imply an intermediate rate of 2.1 GHz. Even reduced, the conceptual intermediate rate is 7.056 MHz, which is why the single combined filter and the polyphase structure are not optimisations but necessities.

```
I, D = 160, 147                         # 48000/44100 reduced by gcd = 300

fc = min(44100, 48000)/2                # = 22050 Hz, the tighter limit, Eq. (11)


# ONE filter serving as both anti-imaging and anti-aliasing

h = I * sig.firwin(3201, fc, fs=44100*I, window="hamming")

out = sig.upfirdn(h, clip, up=I, down=D)      # polyphase: no 7 MHz signal
```

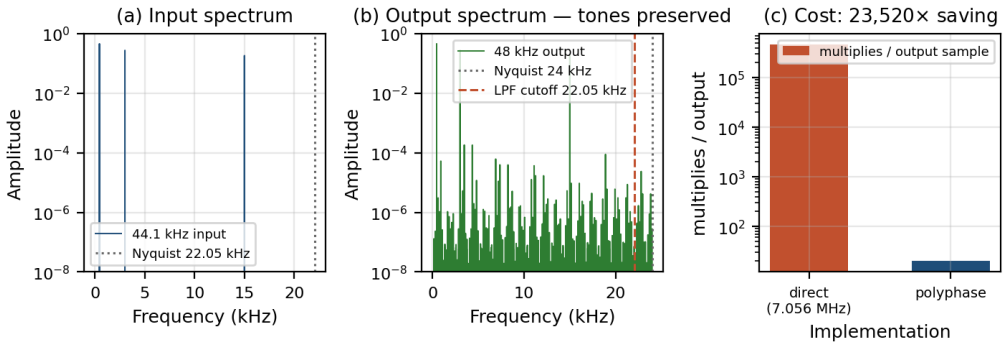


Figure 9. (a) input and (b) output spectra with all tones preserved; (c) arithmetic cost, direct versus polyphase.

Quantity	Value	Note
I / D	160 / 147	gcd(48000, 44100) = 300
Intermediate rate	7.056 MHz	never actually formed
Combined cutoff	22 050 Hz	= min(F_in, F_out)/2
Duration in → out	1.0000 s → 1.0000 s	44 100 → 48 000 samples

Quantity	Value	Note
Tone amplitude error	≤ 0.27 %	440 / 3000 / 15000 Hz
Multiplies per output	470 547 → 20	direct → polyphase
Speed-up	23 520 ×	= I · D, Eq. (12)

Table 6. Fractional resampling results.

The conversion is correct in the sense that matters for audio: the output contains 48 000 samples for one second of input, so the duration and therefore the pitch are unchanged, and all three test tones survive with amplitude errors below 0.27%. The cutoff deserves comment because it is asymmetric between the two directions. Since the output rate is *higher* than the input rate, the binding constraint in (11) is the input Nyquist frequency of 22.05 kHz, not the output's 24 kHz: there is no signal content between them to preserve, and the filter's job is to suppress images rather than to prevent aliasing. Converting the other way, 48 → 44.1 kHz, the same expression would select 22.05 kHz for the opposite reason, and content above it would have to be discarded.

The efficiency figure is the point of the exercise. A direct implementation filters at 7.056 MHz and needs $ND = 3201 \times 147 \approx 4.7 \times 10^5$ multiplications per output sample; the polyphase form computes only the $N/I = 20$ products that are not multiplied by a stuffed zero or thrown away by the down-sampler. The ratio is $I \cdot D =$ **23 520**. At 48 kHz the direct form would demand roughly 2.3×10^{10} multiplications per second, while the polyphase form needs about 10^6 — the difference between a problem no processor will solve in real time and one a microcontroller handles comfortably. The structural insight, not the filter design, is what makes standard-rate audio conversion practical.

5. Conclusion

Multirate processing and adaptive filtering were examined as two answers to the same difficulty: a signal processor must act on information it does not fully control. Decimating a two-tone sequence by 2 without band-limiting first folded the 4 kHz component onto the 1 kHz component in anti-phase, cutting the wanted amplitude from 1.000 to 0.300 = $A_1 - A_2$ and, with equal amplitudes, annihilating the signal entirely to a residual of 1.5×10^{-12} — a demonstration that aliasing damage can consist of energy removed rather than energy added. The anti-aliasing filter restored the tone to 0.9997, and the dual operation showed spectral images at 6, 9, 11 and 14 kHz suppressed to −76.3 dB by an anti-imaging filter carrying the mandatory passband gain of I, without which the amplitude falls by that same factor. An LMS canceller written from scratch reached 44.6 dB against a static notch's 52.5 dB when the mains frequency was exactly known, then held 44.4 dB against the notch's 4.2 dB when the frequency drifted by 0.8 Hz; the accompanying Q study showed that no static design escapes the dilemma, since narrowing the notch to protect the ECG destroys its drift tolerance. The rank-2 structure of a sinusoidal reference explained both why two taps suffice and why the mean-square bound $\mu < 2/\mathrm{tr}(R) = 1.0$ predicted the observed divergence at $\mu = 1.2$ far better than the classical $\mu < 2/\lambda_{max} = 1.91$. Finally, 44.1 to 48 kHz conversion through a single combined filter at 160/147 preserved every test tone to within 0.27% while the polyphase decomposition reduced the arithmetic 23 520-fold. In each case the correct result depended less on the filter itself than on knowing where in the signal chain it had to sit — and, in the adaptive case, on accepting that some of it could not be designed in advance at all.